M2 Mathematics II: Advanced

Level: from undergraduate to graduate (tensors, geometric algebra, differential geometry, groups)

One of eleven documents. Reading order and syllabus: `CHAPTER_ORGANIZATION.md`; plan and status: `ROADMAP.md`.

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Part I TODO(M2/part-I)

1 The geometric (Clifford) algebra

Prerequisites

M1:7 (Multilinear maps, tensors and the tensor algebra); M1:8 (The metric tensor and the musical isomorphisms); M1:9 (The exterior algebra and the wedge product)

The exterior algebra forgets the metric; the geometric algebra puts it back *inside the multiplication*. Instead of $v \otimes v = 0$ we impose $v \otimes v = Q(v)$, the squared length. The remarkable consequence is that the single product ab of two vectors contains both the dot product (its symmetric part) and the wedge product (its antisymmetric part). Let (V, Q) be a quadratic space, i.e. a vector space with a quadratic form Q and its polarization g from (M1:8.1).

1.1 Definition by relations

Definition 1.1 (Clifford algebra)**.** Let $\mathcal{I}_Q \subset T(V)$ be the two-sided ideal generated by $\{v \otimes v - Q(v)\,1 : v \in V\}$. The *Clifford algebra* is $\mathrm{Cl}(V, Q) = T(V)/\mathcal{I}_Q$. Its product is called the *geometric product* and is written by juxtaposition: uv. Thus

$$v\,v = Q(v) = v \cdot v \qquad \text{for every } v \in V. \tag{1.1}$$

When $V = \mathbb{R}^{p,q}$ we write $\mathrm{Cl}(p, q)$; the algebra is then also called the *geometric algebra* $\mathcal{G}(V)$ and its elements *multivectors*.

Theorem 1.2 (Universal property)**.** *Let A be a unital associative algebra and $f : V \to A$ linear with $f(v)^2 = Q(v)1_A$ for all v. Then f extends uniquely to an algebra homomorphism $\mathrm{Cl}(V, Q) \to A$.*

Proof. By Theorem M1:7.11, f extends to $\bar{f} : T(V) \to A$; the hypothesis says $\bar{f}$ kills each generator $v \otimes v - Q(v)$ of $\mathcal{I}_Q$, hence all of $\mathcal{I}_Q$, so $\bar{f}$ descends to the quotient. □

Proposition 1.3 (Fundamental identity)**.** *For $u, v \in V$: $uv + vu = 2\,g(u, v) = 2\,u \cdot v$. In particular, orthogonal vectors anticommute.*

Proof. Polarize (1.1): $(u + v)^2 - u^2 - v^2 = uv + vu$, and by (M1:8.1) this equals $2g(u, v)$. □

Hence, for an orthonormal basis (e_i) with $e_i^2 = \varepsilon_i = \pm 1$, the *entire* algebra is generated by the relations

$$e_i e_j = -e_j e_i \ (i \neq j), \qquad e_i^2 = \varepsilon_i.$$

1.2 Size of the algebra and relation to $\bigwedge V$

Theorem 1.4 (Dimension, basis)**.** $\dim \mathrm{Cl}(V, Q) = 2^n$, *with basis $e_I = e_{i_1} e_{i_2} \cdots e_{i_k}$ for $I = \{i_1 < \cdots < i_k\} \subseteq \{1, \ldots, n\}$ (and $e_\varnothing = 1$).*

Proof sketch. Upper bound. Using $e_ie_j = -e_je_i$ and $e_i^2 = \varepsilon_i$, any word can be reordered into $\pm$ (a scalar) $\cdot\, e_I$ for some I, so the e_I span. *Lower bound (Chevalley's construction).* For $v \in V$ let ι_v be the *interior product* on $\bigwedge V$: the unique antiderivation of degree -1 with $\iota_v(w) = g(v,w)$ for $w \in V$. Define $\rho(v)\,\omega = v \wedge \omega + \iota_v\omega$ on $\bigwedge V$. Using $\iota_v(v\wedge\omega) = g(v,v)\omega - v\wedge\iota_v\omega$ and $\iota_v^2 = 0$, $v\wedge v = 0$ one finds

$$\rho(v)^2\omega = v\wedge\iota_v\omega + \iota_v(v\wedge\omega) = Q(v)\,\omega.$$

By Theorem 1.2, ρ extends to an algebra map $\mathrm{Cl}(V,Q) \to \mathrm{End}(\bigwedge V)$. The map $x \mapsto \rho(x)1$ sends $e_I \mapsto e_{i_1}\wedge\cdots\wedge e_{i_k}$ (for an orthogonal basis the interior terms vanish), which are linearly independent by Theorem M1:9.13. Hence the e_I are independent. □

Corollary 1.5 (Grades). *The map $\bigwedge V \to \mathrm{Cl}(V,Q)$ given by antisymmetrization,*

$$v_1\wedge\cdots\wedge v_k \;\longmapsto\; \frac{1}{k!}\sum_{\sigma\in S_k}\operatorname{sgn}(\sigma)\, v_{\sigma(1)}v_{\sigma(2)}\cdots v_{\sigma(k)},$$

is a (basis-independent) linear isomorphism of vector spaces. We use it to write $\mathcal{G}(V) = \bigoplus_{k=0}^{n}\mathcal{G}^k(V)$ and to define the grade projections $\langle A\rangle_k$. *For* mutually orthogonal *vectors, $a_1a_2\cdots a_k = a_1\wedge\cdots\wedge a_k$. The product does* not *preserve grade, but it preserves the parity (even/odd), so the* even part $\mathcal{G}^+ = \bigoplus_{k\ \text{even}}\mathcal{G}^k$ *is a subalgebra.*

Remark 1.6

So, as vector spaces, $\mathrm{Cl}(V,Q) \cong \bigwedge V$; they differ only in the *multiplication*. When $Q = 0$ the Clifford algebra is the exterior algebra. In this sense the Clifford algebra is a *deformation* of $\bigwedge V$ by the metric.

1.3 Examples

Example 1.7 — Low-dimensional algebras

Write $\mathrm{Cl}(p,q)$ for p basis vectors squaring to $+1$ and q to -1.

- $\mathrm{Cl}(0,1)$: one generator e with $e^2 = -1$; basis $\{1,e\}$; $\mathrm{Cl}(0,1) \cong \mathbb{C}$.
- $\mathrm{Cl}(1,0) \cong \mathbb{R}\oplus\mathbb{R}$ (the split-complex numbers, $e^2 = +1$).
- $\mathrm{Cl}(2,0)$: basis $\{1,e_1,e_2,e_1e_2\}$; $(e_1e_2)^2 = -e_1^2e_2^2 = -1$; $\mathrm{Cl}(2,0) \cong \mathrm{M}_2(\mathbb{R})$.
- $\mathrm{Cl}(0,2)$: with $i = e_1$, $j = e_2$, $k = e_1e_2$ one gets the quaternions: $\mathrm{Cl}(0,2) \cong \mathbb{H}$.
- $\mathrm{Cl}(3,0)$: dimension 8; the Pauli matrices $e_i \mapsto \sigma_i$ give $\mathrm{Cl}(3,0) \cong \mathrm{M}_2(\mathbb{C})$.
- $\mathrm{Cl}(1,3)$ (spacetime algebra, $e_0^2 = +1$, $e_i^2 = -1$) has dimension 16 and is $\cong \mathrm{M}_2(\mathbb{H})$.

1.4 The geometric product of two vectors

Proposition 1.8. *For $a,b \in V$,*

$$ab = a\cdot b + a\wedge b, \qquad a\cdot b = \tfrac{1}{2}(ab+ba), \quad a\wedge b = \tfrac{1}{2}(ab-ba). \tag{1.2}$$

The wedge here coincides with the exterior product under the identification of Corollary 1.5.

Proof. Write $ab = \frac{1}{2}(ab+ba) + \frac{1}{2}(ab-ba)$. The first term is $a\cdot b$ by Proposition 1.3; the second is the antisymmetrization of Corollary 1.5 for $k = 2$. □

Every nonzero vector a with $a^2 \neq 0$ is *invertible*: $a^{-1} = a/a^2$. Consequently equations such as $ax = b$ can be solved by *division*: $x = a^{-1}b$. (Neither the dot product nor the cross product allows this.)

Intuition

Decompose $b = b_{\parallel} + b_{\perp}$ with $b_{\parallel} = (a \cdot b)\, a^{-1}$ the part along a and $b_{\perp}$ the part orthogonal to a. Then $ab_{\parallel} = a \cdot b$ is a scalar (a vector times a parallel vector is a pure number: it is the *projection* information) while $ab_{\perp} = a \wedge b$ is a pure bivector (a vector times a perpendicular vector is an oriented area: it is the *rejection* information). So the geometric product ab is a single object that remembers *how parallel* and *how perpendicular* two vectors are. In the Euclidean case

$$ab = |a||b|\big(\cos\theta + \hat{B}\sin\theta\big), \qquad \hat{B} = \frac{a \wedge b}{|a \wedge b|}, \quad \hat{B}^2 = -1,$$

which is the exact analogue of $e^{i\theta} = \cos\theta + i\sin\theta$ with the imaginary unit replaced by the *plane* spanned by a, b. In the plane, $i \equiv e_1 e_2$ and $i^2 = -1$ appear *naturally*, not as an ad-hoc construction, and in any dimension every oriented plane carries its own "imaginary unit".

2 Operations on multivectors

Prerequisites

M1:9 (The exterior algebra and the wedge product); 1 (The geometric (Clifford) algebra)

The geometric product induces a family of derived operations: grade projections, three involutions, and the inner, outer and contraction products. The left contraction $a \rfloor A$ is the metric incarnation of the interior product ι_a of Section M1:9.6 (Section 5 makes the identification precise).

2.1 Involutions

Definition 2.1. On $\mathcal{G}(V)$ define:

- the *grade involution* $\hat{A}$: the algebra automorphism with $\hat{v} = -v$. On a k-vector, $\hat{A} = (-1)^k A$.
- the *reverse* $\widetilde{A}$: the algebra *anti*-automorphism ($\widetilde{AB} = \widetilde{B}\,\widetilde{A}$) with $\widetilde{v} = v$. On a k-vector, $\widetilde{A} = (-1)^{k(k-1)/2} A$ (reversing the order of the factors in $a_1 \cdots a_k$).
- the *Clifford conjugate* $\bar{A} = \widetilde{\hat{A}}$.

The reverse has signs $+, +, -, -, +, +, \ldots$ for grades $0, 1, 2, 3, 4, 5, \ldots$ In particular bivectors satisfy $\widetilde{B} = -B$.

2.2 Inner, outer and contraction products

Definition 2.2 (Products induced by the geometric product)**.** For homogeneous multivectors $A_r \in \mathcal{G}^r$ and $B_s \in \mathcal{G}^s$:

$$A_r \wedge B_s := \langle A_r B_s \rangle_{r+s} \qquad \text{(outer product)},$$

$$A_r \rfloor B_s := \langle A_r B_s \rangle_{s-r} \qquad \text{(left contraction; } = 0 \text{ if } r > s\text{)},$$
$$A_r * B_s := \langle A_r B_s \rangle_0 \qquad \text{(scalar product)},$$

extended bilinearly to all multivectors. The squared norm (Euclidean case) is $\|A\|^2 = \left\langle A\widetilde{A} \right\rangle_0 \geq 0$.

For a vector a and an r-vector A_r one has the explicit formulas

$$aA_r = a \rfloor A_r + a \wedge A_r, \qquad a \rfloor A_r = \tfrac{1}{2}\big(aA_r - (-1)^r A_r a\big), \quad a \wedge A_r = \tfrac{1}{2}\big(aA_r + (-1)^r A_r a\big), \tag{2.1}$$

of grades $r-1$ and $r+1$ respectively (for $r = 1$ this is (1.2)).

Proposition 2.3 (Properties). *For $a \in V$ and multivectors A, B, C:*

1. *(explicit formula)* $a \rfloor (b_1 \wedge \cdots \wedge b_k) = \sum_{i=1}^{k} (-1)^{i-1} (a \cdot b_i)\, b_1 \wedge \cdots \wedge \widehat{b_i} \wedge \cdots \wedge b_k$.
2. *(antiderivation)* $a \rfloor (A_s \wedge B) = (a \rfloor A_s) \wedge B + (-1)^s A_s \wedge (a \rfloor B)$.
3. *(associativity-type identity)* $A \rfloor (B \rfloor C) = (A \wedge B) \rfloor C$.
4. $(A \wedge B) \wedge C = A \wedge (B \wedge C)$, *and* $A_r \wedge B_s = (-1)^{rs} B_s \wedge A_r$.

Proof idea. (1) and (2) follow by induction from (2.1) (for $k = 2$: $a \rfloor (b \wedge c) = (a \cdot b)c - (a \cdot c)b$). (3) is checked first for $A = a$ a vector, using (2), and then holds by induction on the grade of A. (4) is the statement that the wedge defined here agrees with the exterior product of §M1:9 (Corollary 1.5). A complete proof of (3), on basis blades, is Identity C.4 in the Glossary. □

Example 2.4

$a \rfloor (b \wedge c) = (a \cdot b)\, c - (a \cdot c)\, b$ is a vector lying in the plane $b \wedge c$ and perpendicular to a: indeed $a \cdot \big((a \cdot b)c - (a \cdot c)b\big) = 0$.

Remark 2.5 — Commutator product

For bivectors A, B one has $AB = A \rfloor B + \frac{1}{2}(AB - BA) + A \wedge B$ with the three terms of grades $0, 2, 4$. The antisymmetric middle part, $A \times B := \frac{1}{2}(AB - BA)$ is again a bivector; it makes $\mathcal{G}^2$ a Lie algebra isomorphic to $\mathfrak{so}(p, q)$.

Intuition

- **Wedge = "span".** $A \wedge B$ builds the higher-dimensional element containing both (it is 0 if they share a direction). It is the *join*.
- **Contraction = "the part of B perpendicular to A, measured by A".** $a \rfloor B$ removes the direction a from the blade B and returns what remains inside B, perpendicular to a, with the scale of $|a|$. It lowers grade by one.
- **Geometric product = both at once.** For a vector a and blade B, $aB = a \rfloor B + a \wedge B$ splits into the *lower*-grade part (the part of a lying inside B) and the *higher*-grade part (the part of a sticking out of B). Because the splitting is a sum of different grades, no information is lost, and that is why a can be divided out again.

3 Reflections, rotations and rotors

Prerequisites

1 (The geometric (Clifford) algebra); 2 (Operations on multivectors)

Theorem 3.1 (Reflections). *Let $n \in V$ be invertible ($n^2 \neq 0$). The map $x \mapsto -nxn^{-1}$ is the reflection of V across the hyperplane orthogonal to n.*

Proof. Decompose $x = x_\parallel + x_\perp$, $x_\parallel = (x \cdot n)n^{-1}$ parallel to n, $x_\perp \cdot n = 0$. Then $nx_\parallel = x_\parallel n$ (parallel vectors commute) and $nx_\perp = -x_\perp n$ (Proposition 1.3). Hence $-nxn^{-1} = -x_\parallel + x_\perp$. □

Definition 3.2 (Versors, Pin and Spin). A *versor* is a product $U = v_1 v_2 \cdots v_k$ of invertible vectors. It acts on V by $x \mapsto (-1)^k UxU^{-1}$, which is an orthogonal transformation (a composition of k reflections). The versors with $U\widetilde{U} = \pm 1$ form the group $\mathrm{Pin}(V, Q)$ and the even ones form $\mathrm{Spin}(V, Q) = \mathrm{Pin} \cap \mathcal{G}^+$. An even versor R with $\mathsf{R}\widetilde{\mathsf{R}} = 1$ is called a *rotor*; it acts by the *sandwich product*

$$x \longmapsto \mathsf{R}x\mathsf{R}^{-1} = \mathsf{R}x\widetilde{\mathsf{R}}.$$

By the Cartan–Dieudonné theorem every orthogonal transformation of an n-dimensional space is a product of at most n reflections. Hence every rotation arises from a rotor, and R and $-\mathsf{R}$ give the same rotation (the *double cover* $\mathrm{Spin} \to SO$).

Theorem 3.3 (Rotors in a plane). *Let B be a unit bivector with $B^2 = -1$ (a blade representing an oriented plane in Euclidean space). Then*

$$\mathsf{R} = e^{-\varphi B/2} = \cos\tfrac{\varphi}{2} - B\sin\tfrac{\varphi}{2}$$

acts by $x \mapsto \mathsf{R}x\widetilde{\mathsf{R}}$ as the rotation by angle φ in the plane of B, in the orientation of B, leaving the orthogonal complement of the plane fixed.

Proof. Split $x = x_\parallel + x_\perp$ into the part in the plane of B and the part orthogonal to it. A vector in the plane anticommutes with B (e.g. $e_1e_1e_2 = -e_1e_2e_1$), hence $e^{-\varphi B/2}x_\parallel = x_\parallel e^{\varphi B/2}$ and $\mathsf{R}x_\parallel\widetilde{\mathsf{R}} = x_\parallel e^{\varphi B/2}e^{\varphi B/2} = x_\parallel e^{\varphi B}$. For $B = e_1e_2$ and $x_\parallel = e_1$ this gives $e_1(\cos\varphi + e_1e_2\sin\varphi) = \cos\varphi\, e_1 + \sin\varphi\, e_2$. A vector orthogonal to the plane commutes with B, so $\mathsf{R}x_\perp\widetilde{\mathsf{R}} = x_\perp\mathsf{R}\widetilde{\mathsf{R}} = x_\perp$. □

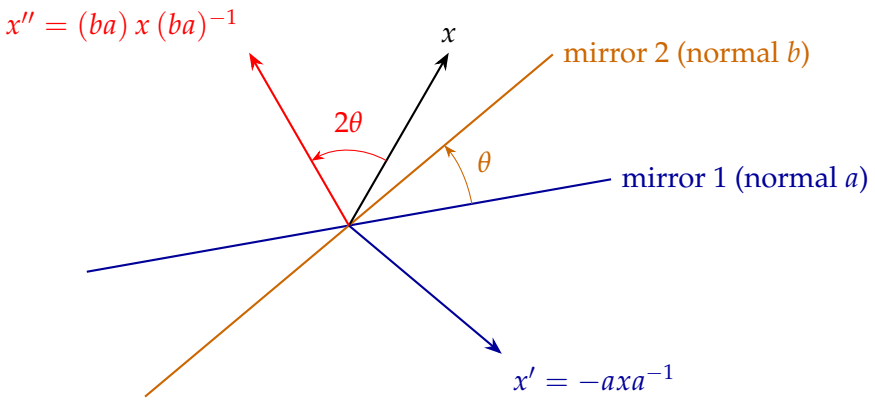


Figure 1: Two reflections compose to a rotation by *twice* the angle θ between the mirrors. The rotor is $\mathsf{R} = ba = e^{-B\theta}$.

Intuition

A reflection is the simplest orthogonal map, and a rotation is just *two reflections in a row* (Fig. 1); this is a classical fact of plane geometry. Algebraically, reflecting first in a and then in b gives $x \mapsto b(-axa^{-1})b^{-1} = (ba)x(ba)^{-1}$, so the *product of the two mirror vectors* $\mathsf{R} = ba$ *is* the rotation. Since the rotation angle is twice the angle between the mirrors, the rotor contains the *half-angle* $\varphi/2$; that is where the half angles of quaternions and spinors come from. The rotor is also dimension-free: no "axis of rotation" is needed (axes exist only in 3D), only the *plane* of rotation, which is the bivector B. Composing rotations is just multiplying rotors: $\mathsf{R}_2\mathsf{R}_1$.

Example 3.4 — Quaternions inside $\mathrm{Cl}(3,0)$

In $\mathcal{G}(\mathbb{R}^3)$ put $i = e_3e_2$, $j = e_1e_3$, $k = e_2e_1$. Then $i^2 = j^2 = k^2 = -1$ and $ij = k$ (direct computation), so the even subalgebra $\mathcal{G}^+(\mathbb{R}^3) = \{\alpha + \text{bivector}\}$ is the quaternion algebra $\mathbb{H}$. Unit quaternions are exactly the rotors of $\mathbb{R}^3$, and $x \mapsto qxq^{-1}$ is the usual quaternion rotation. The pseudoscalar $e_1e_2e_3$ will play the role of the "axis" in §4.

4 Duality: the pseudoscalar, the dual, the cross product and the Hodge star

Prerequisites

M1:8 (The metric tensor and the musical isomorphisms); M1:9 (The exterior algebra and the wedge product); 2 (Operations on multivectors); 3 (Reflections, rotations and rotors)

The pseudoscalar I (the unit n-vector) gives each multivector a perpendicular partner of complementary grade. This single idea explains the cross product, the Hodge star, "axial" vectors and the grad–curl–div pattern met later.

4.1 The pseudoscalar

Let $(e_1, \dots, e_n)$ be an orthonormal basis of a non-degenerate space $V = \mathbb{R}^{p,q}$.

Definition 4.1 (Pseudoscalar)**.** The *unit pseudoscalar* is $I = e_1e_2\cdots e_n = e_1 \wedge \cdots \wedge e_n \in \mathcal{G}^n(V)$. Changing the basis by an orientation-preserving orthogonal map leaves I unchanged; an orientation-reversing one changes its sign. Choosing I is therefore the same as choosing an *orientation* of V.

Proposition 4.2. *1. $I^2 = (-1)^{n(n-1)/2}(-1)^q$ and $I^{-1} = \widetilde{I}/(I\widetilde{I})$. In the Euclidean case $I^{-1} = \widetilde{I} = (-1)^{n(n-1)/2}I$.*

2. $e_iI = (-1)^{n-1}Ie_i$: I commutes with all vectors if n is odd, and anticommutes with them if n is even. In particular I is central in $\mathcal{G}(V)$ when n is odd.

3. For any k-vector A the product AI is a pure $(n-k)$-vector.

Proof. (1) $\widetilde{I} = e_n\cdots e_1 = (-1)^{n(n-1)/2}I$ and $I\widetilde{I} = \prod_i e_i^2 = (-1)^q$, so $I^2 = (-1)^{n(n-1)/2}I\widetilde{I}$. (2) Moving e_i through I it anticommutes with the $n-1$ factors e_j, $j \neq i$, and commutes with e_i. (3) For a basis blade e_J with $|J| = k$, e_JI equals, up to a sign and a product of ε_j, the basis blade e_{J^c} of the complementary indices. □

4.2 The dual

Definition 4.3 (Dual and undual)**.** The *dual* of a multivector A is

$$A^* := A\,I^{-1} \;=\; A \lfloor I^{-1},$$

and the *undual* is $A \mapsto AI$. (The two operations differ from each other only by the scalar factor I^2: $AI = I^2\,A^*$.)

Theorem 4.4 (Properties of the dual)**.** *1. $A \mapsto A^*$ is a linear isomorphism $\mathcal{G}^k(V) \to \mathcal{G}^{n-k}(V)$, and $(A^*)^* = I^{-2}A = \pm A$.*

*2. (*Orthogonal complement*) If A is a k-blade representing the subspace $U \subset V$, then A^* is an $(n-k)$-blade representing the orthogonal complement $U^\perp$ and $\|A^*\| = \|A\|$ (Euclidean case).*

*3. (*Wedge $\leftrightarrow$ contraction*) For all multivectors A, B,*

$$(A \wedge B)^* = A \lfloor B^*, \qquad (A \lfloor B)^* = A \wedge B^*. \tag{4.1}$$

Proof. (1) follows from Proposition 4.2(3) and invertibility of I. (2) Choose an orthonormal basis adapted to U so that (after scaling) $A = e_1 \cdots e_k$. Then $I^{-1} = e_n^{-1} \cdots e_{k+1}^{-1}\, e_k^{-1} \cdots e_1^{-1}$ and, since the $(n-k)$-blade $e_n^{-1} \cdots e_{k+1}^{-1}$ is built from vectors orthogonal to $e_1, \dots, e_k$, it commutes with A up to $(-1)^{k(n-k)}$. Hence $AI^{-1} = (-1)^{k(n-k)} e_n^{-1} \cdots e_{k+1}^{-1}$, a blade spanning $U^\perp$. (3) For homogeneous B we have $B \lfloor I^{-1} = BI^{-1}$ (the product has the single grade $n - \deg B$). Proposition 2.3(3) gives $A \lfloor (BI^{-1}) = A \lfloor (B \lfloor I^{-1}) = (A \wedge B) \lfloor I^{-1} = (A \wedge B)^*$, the first identity. For the second, apply the first to A and $B' = BI$ (so $B'^* = B$): $A \lfloor B = (A \wedge BI)^*$; then $(A \lfloor B)^* = (A \wedge BI)I^{-2} = A \wedge (BI^{-1}) = A \wedge B^*$, using that I^2 is a scalar. □

Corollary 4.5 (Meet and the regressive product)**.** *Define the* meet *(regressive product)*

$$A \vee B := (B^* \wedge A^*)\, I = B^* \lfloor A.$$

If A, B are blades whose subspaces U_A, U_B have $U_A + U_B = V$, then $A \vee B$ is (a multiple of) the blade of $U_A \cap U_B$.

Proof. $U_A^\perp$ and $U_B^\perp$ span $(U_A \cap U_B)^\perp$ when $U_A + U_B = V$, so $B^* \wedge A^*$ represents $(U_A \cap U_B)^\perp$ by Theorem 4.4(2), and undualizing gives $U_A \cap U_B$. The equality of the two expressions follows from (4.1): $(B^* \wedge A^*)I = I^2\,(B^* \wedge A^*)^* = I^2\,B^* \lfloor (A^*)^* = B^* \lfloor A$. □

Example 4.6 — Meet of two planes in $\mathbb{R}^3$

$A = e_1e_2$ and $B = e_2e_3$ (the xy- and yz-planes). With $I = e_1e_2e_3$, $I^{-1} = -I$ one finds $A^* = e_3$, $B^* = e_1$, and $A \vee B = B^* \lfloor A = e_1 \lfloor (e_1e_2) = e_2$: the y-axis, the line of intersection.

4.3 The cross product, rotation axes, and complex numbers

Proposition 4.7 (Cross product)**.** *In $\mathcal{G}(\mathbb{R}^3)$ with $I = e_1e_2e_3$ (so $I^2 = -1$, I central),*

$$a \times b = (a \wedge b)^* = (a \wedge b)I^{-1} = -I\,(a \wedge b), \qquad \textit{equivalently} \qquad a \wedge b = I\,(a \times b).$$

Proof. By bilinearity check on the basis: $e_1e_2I^{-1} = -e_1e_2e_1e_2e_3 = e_3 = e_1 \times e_2$, and cyclically for e_2e_3, e_3e_1. □

Hence *the cross product is the dual of the wedge*. It exists only in $n = 3$ because that is when the dual of a bivector ($n-2$) is a vector (1). It is also why $a \times b$ is a "pseudovector": it changes sign under an orientation reversal, because the dual depends on I.

Example 4.8 — Rotation about an axis

The plane perpendicular to a unit vector $m \in \mathbb{R}^3$, oriented by the right-hand rule, is $B = Im$ (e.g. $Ie_3 = e_1e_2$). Theorem 3.3 gives the rotation by φ about the axis m:

$$\mathsf{R} = \exp\!\left(-\tfrac{\varphi}{2}\, Im\right) = \cos\tfrac{\varphi}{2} - Im\,\sin\tfrac{\varphi}{2}, \qquad x \mapsto \mathsf{R}x\widetilde{\mathsf{R}}.$$

In 2D, $I = e_1e_2$ acts as the imaginary unit and the dual of a vector is that vector turned by a quarter turn.

4.4 Relation to the Hodge star and to the dual space

Let V be Euclidean and oriented, with the inner product on $\bigwedge^k V$ for which $\{e_I\}$ is orthonormal; in terms of the geometric product, $\langle A, B\rangle = \left\langle \widetilde{A}B \right\rangle_0$.

Definition 4.9 (Hodge star)**.** The *Hodge star* $\star : \bigwedge^k V \to \bigwedge^{n-k} V$ is defined by

$$\alpha \wedge \star\beta = \langle \alpha, \beta\rangle\, I \qquad \text{for all } \alpha, \beta \in \textstyle\bigwedge^k V.$$

Theorem 4.10. *For a k-vector A in the Euclidean case,*

$$\star A = \widetilde{A}\, I = (-1)^{k(k-1)/2 + n(n-1)/2}\, A^*.$$

So the Hodge star and the geometric-algebra dual agree up to a grade-dependent sign. Moreover $\star\star = (-1)^{k(n-k)}$ on k-vectors.

Proof. For an orthonormal basis blade e_J ($|J| = k$) we have $\widetilde{e_J} = e_J^{-1}$. Then $e_J \wedge (\widetilde{e_J} I) = e_J\, e_J^{-1} I = I$, since the product of e_J with the complementary blade $e_J^{-1} I$ (built from the other basis vectors) is a wedge; and $e_{J'} \wedge (\widetilde{e_J} I) = 0$ for $J' \neq J$ because the indices overlap. This is precisely the defining property of $\star e_J$. The sign relation follows from $\widetilde{A} = (-1)^{k(k-1)/2} A$ and $I^{-1} = (-1)^{n(n-1)/2} I$. The relation $\star\star = (-1)^{k(n-k)}$ is proved in Identity C.7 of the Glossary. □

Remark 4.11 — The "other" dual

The word *dual* has two meanings that are connected here. The *dual space* V^* is intrinsically defined, whereas $V \cong V^*$ needs the metric ($\flat$). Intrinsically, the wedge pairing $\bigwedge^k V \times \bigwedge^{n-k} V \to \bigwedge^n V \cong \mathbb{K}$ is perfect, so $\bigwedge^{n-k} V \cong (\bigwedge^k V)^*$ once a basis (orientation and volume) I of $\bigwedge^n V$ is chosen. Combined with the metric identification $(\bigwedge^k V)^* \cong \bigwedge^k V$, this is exactly the map $A \mapsto A^*$ (or $\star A$). The pseudoscalar I packages *both* the orientation and the metric volume.

Intuition

- **Dual = orthogonal complement.** Every subspace has a perpendicular partner: a plane in 3D has a normal line, a line has a normal plane. The dual swaps a k-dimensional oriented subspace with its $(n-k)$-dimensional partner (and keeps the magnitude). Describing a plane *by its normal* or *by its tangent bivector* are two views of the same object; the dual converts between them (for example $e_1 \wedge e_2 \leftrightarrow e_3$ in $\mathbb{R}^3$).
- **Wedge $\leftrightarrow$ meet.** The wedge joins subspaces (spans). In the dual picture it becomes a contraction/meet (intersections). The identity $(A \wedge B)^* = A \rfloor B^*$ says that "the complement of a span is what is left perpendicular to both". Intersections of planes, lines and so on, without coordinates.

- **The pseudoscalar is the orientation of space.** Multiplying by I^{-1} is a "quarter-turn of the whole space": in 2D it is multiplication by $\pm i$; in 3D it turns planes into axes. Reversing the orientation flips the sign of the dual; this is why the cross product is a *pseudo*vector and the magnetic field B is better treated as a *bivector*.

The Hodge star on forms. The definition above makes sense for any oriented space with an inner product, so it applies verbatim to V^* with the inner product induced by the metric (g^{ij}, Section M1:8): for k-forms, $\alpha \wedge \star\beta = \langle \alpha, \beta \rangle$ vol, where vol $= e^1 \wedge \cdots \wedge e^n$ for a positively oriented orthonormal basis. Under the identification $e^i \leftrightarrow e_i$ of an orthonormal basis, $\star$ on forms is the $\star$ on multivectors of Theorem 4.10; on a manifold it acts pointwise on $\bigwedge^k T_p^* M$. In $\mathbb{R}^3$, for example, $\star dx = dy \wedge dz$, $\star dy = dz \wedge dx$, $\star dz = dx \wedge dy$ and $\star 1 = dx \wedge dy \wedge dz$.

5 Synthesis I: contraction, duality and the role of the metric

Prerequisites

M1:6 (Vectors, covectors and contraction); M1:8 (The metric tensor and the musical isomorphisms); M1:9 (The exterior algebra and the wedge product); M1:16 (The Gauss (divergence) theorem); 2 (Operations on multivectors); 4 (Duality: the pseudoscalar, the dual, the cross product and the Hodge star)

Synthesis: what this section does

Three languages have now appeared for the same linear algebra: *index* (tensor) notation, *forms* on V^*, and *multivectors* in V. This section lines them up. It answers three questions: which of the "contractions" are really the same operation (Section 5.1), which operations genuinely need a metric (Section 5.2), and how the conventions of the different sources are reconciled (Section 5.3).

5.1 Contraction in four guises

The word "contraction" has been used four times; they are one idea at four levels of structure.

Guise	Acts on	Formula	Metric?
Evaluation $\langle \omega, v \rangle$	$V^* \times V \to \mathbb{R}$	$\omega_i v^i$	no
Tensor contraction	$T^\ell_k \to T^{\ell-1}_{k-1}$	trace $A^i{}_i$; $f_i v^i$; Ricci $R^i{}_{ijk}$	no
Interior product ι_v	k-forms $\to$ $(k-1)$-forms	$(\iota_v \alpha)(w_\bullet) = \alpha(v, w_\bullet)$	no
Left contraction $a \lrcorner A$	k-vectors $\to$ $(k-1)$-vectors	$a \lrcorner (b \wedge c) = (a \cdot b)c - (a \cdot c)b$	yes ($\cdot$ is g)

The first three need only V and V^*. The fourth acts on vectors and needs a metric, but it is the *same* operation as the third once a metric is available to translate between V and V^*:

Proposition 5.1 (left contraction = interior product under $\flat$)**.** *Extend $\flat : V \to V^*$ to an algebra homomorphism of exterior algebras, $(v_1 \wedge \cdots \wedge v_k)^\flat = v_1^\flat \wedge \cdots \wedge v_k^\flat$. Then for every $a \in V$ and every multivector A,*

$$(a \lrcorner A)^\flat = \iota_a (A^\flat).$$

Proof. Both sides are antiderivations in A of degree -1 (for $a\rfloor$, see the properties in Section 2; for ι_a, (M1:9.11)). On a vector b they agree: $(a \rfloor b)^\flat = (a \cdot b) = g(a,b) = b^\flat(a) = \iota_a b^\flat$. Since multivectors are sums of wedges of vectors, an induction on the grade using the antiderivation rule shows that they agree everywhere. □

Example 5.2 — the three-vector check

Take $a, b, c \in \mathbb{R}^3$ with the Euclidean metric. Then $(a \rfloor (b \wedge c))^\flat = \big((a\cdot b)c - (a\cdot c)b\big)^\flat = (a\cdot b)c^\flat - (a\cdot c)b^\flat$, and $\iota_a(b^\flat \wedge c^\flat) = b^\flat(a)\, c^\flat - c^\flat(a)\, b^\flat$ by (M1:9.2); the two agree because $b^\flat(a) = a \cdot b$. In coordinates this is exactly the planar computation of Example M1:9.19: $\iota_v(\mathrm{d}x \wedge \mathrm{d}y) = v^x \mathrm{d}y - v^y \mathrm{d}x$.

Synthesis: the same identities, three notations

Fact	Forms	Multivectors
antiderivation	$\iota_v(\alpha \wedge \beta) = \iota_v\alpha \wedge \beta + (-1)^k \alpha \wedge \iota_v\beta$	$a \rfloor (A_k \wedge B) = (a \rfloor A_k) \wedge B + (-1)^k A_k \wedge (a \rfloor B)$
nilpotent	$\iota_v \iota_v = 0$	$a \rfloor (a \rfloor A) = (a \wedge a) \rfloor A = 0$
wedge is not metric	$\alpha \wedge \beta$	$A \wedge B$ (grade part of AB, but defined without g)

5.2 Which operations need a metric?

The divisions drawn in Sections M1:8–4 can be gathered into one table; each row is justified where the object is introduced.

Operation / object	Metric?	What it needs instead
$\langle \omega, v \rangle$, tensor contraction, $\otimes$	no	V, V^*
$\wedge$, ι_v, $\bigwedge V$, pull-back ϕ^*	no	V, V^*
d, $\mathcal{L}_X$, integration of forms, Stokes	no	smooth structure (and orientation for $\int$)
$\mathrm{div}_\mu X$ (Section M1:16)	no	a volume form μ (a metric supplies one)
$\flat$, $\sharp$, $\nabla f = (\mathrm{d}f)^\sharp$	yes	g
Hodge star $\star$, dual $A^* = AI^{-1}$	yes	g and an orientation
$a \times b$, curl	yes	g, orientation, and $n = 3$
Geometric product ab, $a \rfloor A$, rotors	yes	Q (the metric)
Levi-Civita connection, geodesics, curvature	yes	g (a bare connection needs only the manifold)

Remark 5.3 — dual basis versus reciprocal frame

The *dual basis* $e^i \in V^*$ is defined without a metric by $e^i(e_j) = \delta^i{}_j$. The *reciprocal frame* of geometric algebra is the vector $e^i_\sharp = g^{ij} e_j \in V$ with $e^i_\sharp \cdot e_j = \delta^i{}_j$; it is the dual basis transported by $\sharp$. For an orthonormal frame with $e_i^2 = \varepsilon_i = \pm 1$ it is simply $e^i_\sharp = \varepsilon_i e_i$. used by the vector

derivative $\boldsymbol{\nabla} = e^i_\sharp\,\partial_i$ in Section 8.

5.3 A dictionary of conventions

Because the three sources were written independently, the same object appears with different normalisations. They have been unified as follows; this is the place to look when a formula elsewhere in the literature seems to disagree by a factor or a sign.

Remark 5.4 — the factor $k!$ between the two realisations of $\wedge$

Let $\alpha^1,\dots,\alpha^k \in V^*$ and $v_1,\dots,v_k \in V$. The form convention of Section M1:9 gives $(\alpha^1\wedge\cdots\wedge\alpha^k)(v_1,\dots,v_k) = \det[\alpha^i(v_j)]$. If instead one embeds multivectors in $T^k(V)$ by $v_1\wedge\cdots\wedge v_k \mapsto \frac{1}{k!}\sum_\sigma \operatorname{sgn}(\sigma)\, v_{\sigma(1)}\otimes\cdots\otimes v_{\sigma(k)}$ and evaluates the resulting tensor on $(\alpha^1,\dots,\alpha^k)$, one obtains $\frac{1}{k!}\det[\alpha^i(v_j)]$. The two realisations are identified by multiplying degree-k elements by $k!$: this map is an algebra isomorphism, because the factor $\frac{(k+l)!}{k!\,l!}$ in (M1:9.6) satisfies $(k+l)! = k!\,l!\cdot\frac{(k+l)!}{k!\,l!}$. Nothing in the abstract algebra changes; only the numbers one gets by evaluating on tensors do.

Item	Convention in this document	Other conventions met in the sources
Signature (p,q)	p positive, q negative; Euclidean $(n,0)$ (pitfall in Section M1:8.1)	GR notes: (negative, positive); mostly minus $(1,3)$ in the spacetime algebra
Wedge, forms	$\mathrm{d}x\wedge\mathrm{d}y = \mathrm{d}x\otimes\mathrm{d}y - \mathrm{d}y\otimes\mathrm{d}x$; det pairing	Kobayashi–Nomizu: factor $\frac{1}{2}$
Wedge, multivectors	$v\wedge w \mapsto \frac{1}{2}(v\otimes w - w\otimes v)$ in $T(V)$; $a\wedge b = \frac{1}{2}(ab-ba)$	see Remark 5.4
Tensor type (ℓ,k)	ℓ contravariant (up), k covariant (down)	some texts reverse the order
Dual basis	e^i ($=\mathrm{d}x^i$ in coordinates)	e^{*i} in the GR notes
Hodge star vs. dual	$\star A = \widetilde{A}I = (-1)^{k(k-1)/2+n(n-1)/2}A^*$	dual with I or I^{-1} on either side
Curvature	$R(X,Y)Z = \nabla_X\nabla_Y Z - \nabla_Y\nabla_X Z - \nabla_{[X,Y]}Z$	Carroll/MTW [21, 10] agree; others differ in index order (Section 12)
∇	plain: covariant derivative. Bold $\boldsymbol{\nabla}$: vector derivative	Hestenes [22] writes ∇ for $\boldsymbol{\nabla}$

Key points of Part I

- $T(V)$ is the free algebra; the exterior algebra ($v^2=0$) is metric-free and carries forms; the Clifford algebra ($v^2 = Q(v)$) contains the metric and carries rotations.
- The four "contractions" coincide under $\flat$: $(a\rfloor A)^\flat = \iota_a(A^\flat)$.
- Duality $A^* = AI^{-1}$ is the geometric-algebra face of the Hodge star; the cross product is $a\times b = (a\wedge b)^*$ and exists only for $n=3$.

6 Groups: symmetry, finite groups and representations

TODO

TODO(M2/ch06): this chapter has no content yet. Its place in the reading order is listed in `CHAPTER_ORGANIZATION.md`.

Prerequisites: M1:8 (The metric tensor and the musical isomorphisms); 3 (Reflections, rotations and rotors)

Part II TODO(M2/part-II)

7 The Lie derivative and Cartan's formula

Prerequisites

M1:9 (The exterior algebra and the wedge product); M1:12 (Tensor fields and differential forms); M1:16 (The Gauss (divergence) theorem)

How does a field change from point to point? For a function the question is easy; for vectors and forms we must compare objects living in different vector spaces.

Intuition: the problem

For a function f it is easy to compare $f(p)$ and $f(q)$ at nearby points: they are both numbers. For a vector or tensor field, $v_p \in T_pM$ and $v_q \in T_qM$ live in *different* vector spaces: to subtract them we need a rule to **identify** T_pM **with** T_qM. There are two natural rules, giving two derivatives:

	identification of $T_pM \to T_qM$	extra data needed
Lie derivative $\mathcal{L}_X$	differential of the flow of X	none (but depends on X *near* p)
Covariant derivative ∇_X	parallel transport along the flow line	a *connection* ∇

Remark 7.1 — a third derivative

A differential form admits a third derivative, the exterior derivative d (Section M1:12.3), which needs *no identification of tangent spaces at all*: it uses only the antisymmetrised partial derivatives, and is therefore independent of both a vector field (unlike $\mathcal{L}_X$) and a connection (unlike ∇). Cartan's formula below relates it to the Lie derivative, and Section 14 compares all three.

7.1 Vector fields and their flows

Formal idea: flow of a vector field

A vector field $X \in \mathfrak{X}(M)$ is equivalent to an ODE for curves: $\dot\gamma(t) = X_{\gamma(t)}$, *i.e.* in a chart $\dot x^i = X^i(x)$. By existence and uniqueness of ODE solutions X defines a *flow* ϕ_t (defined for small t near each point): $\phi_t(x)$ is the solution at time t with initial condition x. Properties: $\phi_0 = \mathrm{id}$, $\phi_{t_1} \circ \phi_{t_2} = \phi_{t_1+t_2}$ (where defined), $\phi_t^{-1} = \phi_{-t}$, $\frac{\mathrm{d}}{\mathrm{d}t}\phi_t(x) = X_{\phi_t(x)}$, and to first order $\phi_\varepsilon(x) = x + \varepsilon X(x) + O(\varepsilon^2)$.

Intuition: wind blowing a leaf

X is a stationary wind map; ϕ_t moves every point for time t following the wind. The family $t \mapsto \phi_t$ is a one-parameter group of local diffeomorphisms: X is its *infinitesimal generator*.

7.2 Definition and properties

Formal idea: Lie derivative

For $\tau \in \Pi^\ell_k(M)$ and $X \in \mathfrak{X}(M)$ with flow ϕ_ε,

$$(\mathcal{L}_X \tau)_p = \frac{\mathrm{d}}{\mathrm{d}\varepsilon}\Big|_{\varepsilon=0} (\phi_\varepsilon{}^*\tau)_p = \lim_{\varepsilon\to 0} \frac{(\phi_\varepsilon{}^*\tau)_p - \tau_p}{\varepsilon} \in \mathcal{T}^\ell_k(T_pM),$$

i.e. $(\phi_\varepsilon{}^*\tau)_p = \tau_p + \varepsilon(\mathcal{L}_X\tau)_p + O(\varepsilon^2)$. Properties:

- $\mathcal{L}_X f = X(f)$ for functions.
- (Leibniz) $\mathcal{L}_X(\tau_1 \otimes \tau_2) = \mathcal{L}_X\tau_1 \otimes \tau_2 + \tau_1 \otimes \mathcal{L}_X\tau_2$; it "differentiates every slot": $\mathcal{L}_X(\tau(v_1,\dots)) = (\mathcal{L}_X\tau)(v_1,\dots) + \tau(\mathcal{L}_X v_1,\dots) + \dots$
- For vector fields, $\mathcal{L}_X Y = [X,Y] = X\circ Y - Y \circ X$ (the *Lie bracket*), *i.e.* $[X,Y]f = X(Yf) - Y(Xf)$.
- $\mathcal{L}_X(\mathrm{d}f) = \mathrm{d}(\mathcal{L}_X f)$.
- In coordinates $X = X^j\partial_j$: $\mathcal{L}_X\partial_i = -\partial_i X^j\,\partial_j$, $\mathcal{L}_X \mathrm{d}x^i = \partial_j X^i\,\mathrm{d}x^j$, and for a metric

$$(\mathcal{L}_X g)_{ab} = X^k\partial_k g_{ab} + g_{kb}\,\partial_a X^k + g_{ak}\,\partial_b X^k. \tag{7.1}$$

Intuition: "how does τ change when I drag it with the flow, compared with the original?"

You push τ along the flow of X for a tiny time (like a sheet of rubber being dragged by the water) and compare with what τ actually is there. $\mathcal{L}_X\tau = 0$ means τ is **invariant** under the flow: it is a symmetry. The bracket $[X,Y]$ measures the failure of the flows of X and Y to commute: go ε along X, then ε along Y, then back along X and Y: you end up displaced by $\varepsilon^2[X,Y]$ (a rectangle that does not close).

Careful: $\mathcal{L}_X$ is *not* a connection

$\mathcal{L}_{fX}Y = f\,\mathcal{L}_X Y - Y(f)\,X \neq f\,\mathcal{L}_X Y$. The Lie derivative needs the flow of X near the point (the derivatives of X appear in it, *e.g.* $\partial_a X^k$ in (7.1)), not just the value X_p. A covariant derivative, in contrast, is $C^\infty(M)$-linear in X.

Example 7.2 — a Lie bracket

On $\mathbb{R}^2$ take $X = \partial_x$, $Y = x\,\partial_y$. For any f: $[X,Y]f = \partial_x(x\,\partial_y f) - x\,\partial_y\partial_x f = \partial_y f$, so $[\partial_x, x\partial_y] = \partial_y \neq 0$: moving first in x and then in y is not the same as in the opposite order because the y-speed depends on x.

Formal idea: invariant tensors and Killing fields

τ is *invariant under* X if $\mathcal{L}_X\tau = 0$, equivalently ${\phi_t}^*\tau = \tau$ for all t. If a metric satisfies $\mathcal{L}_X g = 0$, then X is called an *infinitesimal isometry* or **Killing vector field**; its flow consists of (local) isometries.

Example 7.3 — Killing field and conformal Killing field of the plane in polar coordinates

With $g = \mathrm{diag}(1, r^2)$ in polar coordinates and formula (7.1):

- $X = \partial_\varphi$ (rotations, $X^r = 0$, $X^\varphi = 1$): all $\partial_a X^k = 0$ and $X^k\partial_k g_{ab} = \partial_\varphi g_{ab} = 0$, hence $\mathcal{L}_X g = 0$: **Killing**. (Its flow is the rotation $\phi_\varepsilon(r,\varphi) = (r, \varphi+\varepsilon)$, which clearly preserves lengths.)
- $X = r\,\partial_r$ (dilations $\phi_\varepsilon(x) = e^\varepsilon x$): $(\mathcal{L}_X g)_{rr} = r\partial_r(1) + 2g_{rr}\partial_r X^r = 2$, $(\mathcal{L}_X g)_{\varphi\varphi} = r\partial_r(r^2) = 2r^2$, so $\mathcal{L}_X g = 2g$: not Killing but *conformal* (lengths are rescaled by $e^{2\varepsilon}$: ${\phi_\varepsilon}^* g = e^{2\varepsilon} g$). ✓

Intuition: why Killing fields matter in GR

Symmetries of spacetime give conserved quantities: a time-translation Killing field gives energy conservation, a rotation Killing field gives angular momentum (Section 11). In the Schwarzschild solution ∂_t and ∂_φ are Killing.

7.3 Cartan's magic formula

Theorem 7.4 (Cartan). *For every differential form ω and vector field X,*

$$\mathcal{L}_X\omega = \mathrm{d}\,\iota_X\omega + \iota_X\,\mathrm{d}\omega. \tag{7.2}$$

Proof. **Step 1 (both sides are derivations).** $\mathcal{L}_X$ is a derivation of $\wedge$ (Leibniz rule for the flow derivative). The right-hand side is the graded commutator $[\mathrm{d}, \iota_X]$ of two antiderivations ((M1:9.11), (M1:9.4)), hence a derivation of degree 0. **Step 2 (they commute with** d**).** $\mathcal{L}_X\mathrm{d} = \mathrm{d}\mathcal{L}_X$ (pull-back commutes with d), and $\mathrm{d}[\mathrm{d},\iota_X] = \mathrm{d}\iota_X\mathrm{d} = [\mathrm{d},\iota_X]\mathrm{d}$ because $\mathrm{dd} = 0$. **Step 3 (they agree on functions).** $\mathrm{d}\iota_X f + \iota_X\mathrm{d}f = 0 + X(f) = \mathcal{L}_X f$. **Step 4 (conclusion).** Locally every form is a sum of terms $f\,\mathrm{d}g_1\wedge\cdots\wedge\mathrm{d}g_k$; two degree-0 derivations commuting with d and agreeing on functions agree on f and on $\mathrm{d}g_i = \mathrm{d}(g_i)$, hence on all forms. □

Example 7.5 — Cartan's formula for a 1-form

Take $\omega = x\,\mathrm{d}y$ on $\mathbb{R}^2$ and $X = \partial_x$ (flow $\phi_t(x,y) = (x+t, y)$). Then $\iota_X\omega = x\,\mathrm{d}y(\partial_x) = 0$, so $\mathrm{d}\iota_X\omega = 0$; $\mathrm{d}\omega = \mathrm{d}x\wedge\mathrm{d}y$ and $\iota_X\mathrm{d}\omega = \mathrm{d}y$. Hence $\mathrm{d}\iota_X\omega + \iota_X\mathrm{d}\omega = \mathrm{d}y$. Directly from the flow: ${\phi_t}^*\omega = (x+t)\,\mathrm{d}y$ and $\frac{\mathrm{d}}{\mathrm{d}t}\big|_0 {\phi_t}^*\omega = \mathrm{d}y$ ✓. In components, for $\omega = \omega_i\mathrm{d}x^i$: $\mathcal{L}_X\omega = (X^j\partial_j\omega_i + \omega_j\partial_i X^j)\,\mathrm{d}x^i$, and expanding $\mathrm{d}(\omega_j X^j) + \iota_X(\partial_j\omega_i\,\mathrm{d}x^j\wedge\mathrm{d}x^i)$ gives the same expression.

Remark 7.6 — Cartan's formula is the engine of the integral theorems

Since $\mathrm{d}\mu = 0$ for a top-degree form, (7.2) gives $\mathcal{L}_X\mu = \mathrm{d}(\iota_X\mu)$: the rate of change of volume along a flow is the exterior derivative of the flux form $\iota_X\mu$. This is precisely the divergence theorem in disguise (Section M1:16); and for fluids it is the step that turns Euler's equation into the exact form $(\partial_t + \mathcal{L}_u)u^\flat = \mathrm{d}(\cdots)$ behind Kelvin's circulation theorem (Section F1:5.1).

7.4 Operator relations: $[\partial_i, x^j]$ and $\{\iota_i, \mathrm{d}x^j\wedge\}$

Let x^j act on forms by multiplication, ∂_i by differentiating the coefficients, $e^j = \mathrm{d}x^j\wedge$ by wedging, and $\iota_i = \iota_{\partial_i}$ by contracting.

Proposition 7.7. *For every form α and function f:*

$$\partial_i(x^j f) - x^j\partial_i f = \delta_i^j f \qquad \textit{i.e. } [\partial_i, x^j] = \delta_i^j, \tag{7.3}$$

$$\iota_i(\mathrm{d}x^j\wedge\alpha) + \mathrm{d}x^j\wedge\iota_i\alpha = \delta_i^j\alpha \qquad \textit{i.e. } \{\iota_i, \mathrm{d}x^j\wedge\} = \delta_i^j, \tag{7.4}$$

$$\mathrm{d}(x^j\alpha) - x^j\mathrm{d}\alpha = \mathrm{d}x^j\wedge\alpha \qquad \textit{i.e. } [\mathrm{d}, x^j] = \mathrm{d}x^j\wedge, \tag{7.5}$$

$$\mathrm{d}(\alpha_I\,\mathrm{d}x^I) = \partial_i\alpha_I\,\mathrm{d}x^i\wedge\mathrm{d}x^I \qquad \textit{i.e. } \mathrm{d} = \mathrm{d}x^i\wedge\partial_i \textit{ on coefficients}, \tag{7.6}$$

$$\mathcal{L}_{\partial_i}(\alpha_I\,\mathrm{d}x^I) = (\mathrm{d}\iota_i + \iota_i\mathrm{d})(\alpha_I\mathrm{d}x^I) = \partial_i\alpha_I\,\mathrm{d}x^I. \tag{7.7}$$

Proof. (7.3): Leibniz rule, $\partial_i(x^j f) = \delta_i^j f + x^j\partial_i f$. (7.4): the antiderivation rule (M1:9.11) with the 1-form $\mathrm{d}x^j$ gives $\iota_i(\mathrm{d}x^j\wedge\alpha) = (\iota_i\mathrm{d}x^j)\alpha - \mathrm{d}x^j\wedge\iota_i\alpha = \delta_i^j\alpha - \mathrm{d}x^j\wedge\iota_i\alpha$. (7.5): Leibniz rule (M1:9.4) with a 0-form, $\mathrm{d}(x^j\alpha) = \mathrm{d}x^j\wedge\alpha + x^j\mathrm{d}\alpha$. (7.6) is the definition of d in coordinates. For (7.7): $\mathcal{L}_{\partial_i}\mathrm{d}x^j = \mathrm{d}(\iota_i\mathrm{d}x^j) + \iota_i\mathrm{d}\mathrm{d}x^j = \mathrm{d}(\delta_i^j) + 0 = 0$, and $\mathcal{L}_{\partial_i}$ is a derivation, so only the coefficient is differentiated. □

Numerical check. Check of (7.4) in $\mathbb{R}^2$ with $\alpha = x\,\mathrm{d}y$ and $i = j = x$: $\mathrm{d}x\wedge\alpha = x\,\mathrm{d}x\wedge\mathrm{d}y$ and $\iota_{\partial_x}(x\,\mathrm{d}x\wedge\mathrm{d}y) = x\,\mathrm{d}y$; $\iota_{\partial_x}\alpha = x\,\mathrm{d}y(\partial_x) = 0$. The sum is $x\,\mathrm{d}y = \alpha$ ✓. Check of (7.7) with $\alpha = f\,\mathrm{d}x\wedge\mathrm{d}y$: $\mathrm{d}\alpha = 0$ (a 3-form in the plane), so $\mathcal{L}_{\partial_x}\alpha = \mathrm{d}(f\,\mathrm{d}y) = f_x\,\mathrm{d}x\wedge\mathrm{d}y$ ✓.

Remark 7.8 — Heisenberg and the fermions

(7.3) and (7.4) are the same duality $\mathrm{d}x^j(\partial_i) = \delta_i^j$ written for the even pair (x^j, ∂_i) with a commutator and for the odd pair $(\mathrm{d}x^j\wedge, \iota_i)$ with an anticommutator. The first is the Heisenberg relation of quantum mechanics ($p = -i\hbar\partial_x$, $[x, p] = i\hbar$); the second is the canonical anticommutation relation of fermions, and the exterior algebra $\bigwedge(V^*)$ is the corresponding Fock space. In both cases ∂_x (resp. ι_{∂_x}) is the "annihilation" partner of x (resp. $\mathrm{d}x\wedge$).

Key points of manifolds, forms and the Lie derivative

- x is a function, $\mathrm{d}x = \mathrm{d}(x)$ a covector, $\partial/\partial x$ the dual vector, with $\mathrm{d}x(\partial_x) = 1$. $\mathrm{d}x$ depends only on the function x; $\partial/\partial x$ depends on the whole chart.
- $\partial/\partial x = \iota_{\partial_x}\mathrm{d}$ on functions; $\mathrm{d} = \mathrm{d}x^i\wedge\partial_i$ on forms; and $[\partial_i, x^j] = \{\iota_i, \mathrm{d}x^j\wedge\} = \delta_i^j$.
- The three derivatives of geometry: d (forms; no extra data), $\mathcal{L}_X$ (needs a vector field), ∇ (needs a connection); they are tied by $\mathcal{L}_X = \mathrm{d}\iota_X + \iota_X\mathrm{d}$ on forms.

8 The exterior derivative in geometric algebra

Prerequisites

M1:12 (Tensor fields and differential forms); 2 (Operations on multivectors); 4 (Duality: the pseudoscalar, the dual, the cross product and the Hodge star); 5 (Synthesis I: contraction, duality and the role of the metric)

The exterior derivative now returns in a second language. The explicit coordinate formula (M1:12.1) is the "wedge half" of one single first-order operator on multivector fields, the *vector derivative* $\boldsymbol{\nabla} = e^i_\sharp\,\partial_i$ (bold nabla, not to be confused with the covariant derivative ∇). Differential forms are identified with multivector fields through $\flat$ (Proposition 5.1). Let $V = \mathbb{R}^{p,q}$ with an orthonormal frame (e_i) and reciprocal frame (e^i), $e^i \cdot e_j = \delta^i_j$ (so $e^i = \varepsilon_i e_i$). In this section e^i always denotes this reciprocal frame, i.e. the *vector* $e^i_\sharp$ of Remark 5.3, and not the dual basis covector. On V itself, coordinates x^i are defined by $x = x^i e_i$. Using the metric, identify a differential form with a multivector field via $\mathrm{d}x^i \leftrightarrow e^i$:

$$\omega = \sum_I \omega_I\,\mathrm{d}x^I \quad\longleftrightarrow\quad F = \sum_I \omega_I\, e^{i_1}\wedge\cdots\wedge e^{i_k}.$$

Definition 8.1 (Vector derivative)**.** The *vector derivative* (Dirac operator) is $\boldsymbol{\nabla} = e^i\partial_i$. Its action on a multivector field F is its geometric product with F:

$$\boldsymbol{\nabla}F = e^i\,\partial_i F = \boldsymbol{\nabla}\rfloor F + \boldsymbol{\nabla}\wedge F, \qquad \boldsymbol{\nabla}\rfloor F := e^i\rfloor\partial_i F, \quad \boldsymbol{\nabla}\wedge F := e^i\wedge\partial_i F.$$

For a k-vector field F, $\boldsymbol{\nabla}\wedge F$ has grade $k+1$ ("curl-like") and $\boldsymbol{\nabla}\rfloor F$ has grade $k-1$ ("divergence-like"); this is formula (2.1) with the vector a replaced by the operator $\boldsymbol{\nabla}$.

Theorem 8.2 (d is $\boldsymbol{\nabla}\wedge$)**.** *Under the identification above:*

1. $\mathrm{d}\omega \;\longleftrightarrow\; \boldsymbol{\nabla}\wedge F$.
2. *(Euclidean case) the codifferential* $\delta = (-1)^{n(k+1)+1}\star\,\mathrm{d}\star$ *corresponds to* $-\boldsymbol{\nabla}\rfloor F$*; hence* $\boldsymbol{\nabla} = \mathrm{d} - \delta$ *acting on forms.*
3. $\boldsymbol{\nabla}\wedge(\boldsymbol{\nabla}\wedge F) = 0$ *and* $\boldsymbol{\nabla}\rfloor(\boldsymbol{\nabla}\rfloor F) = 0$.
4. $\boldsymbol{\nabla}^2 = \boldsymbol{\nabla}\cdot\boldsymbol{\nabla} = \sum_i \varepsilon_i\partial_i^2$ *(the Laplacian, or d'Alembertian), and* $\boldsymbol{\nabla}^2 = \boldsymbol{\nabla}\rfloor(\boldsymbol{\nabla}\wedge\,\cdot\,) + \boldsymbol{\nabla}\wedge(\boldsymbol{\nabla}\rfloor\,\cdot\,) = -(\mathrm{d}\delta + \delta\mathrm{d})$ *in the Euclidean case.*
5. *(Duality) in terms of the dual of Definition 4.3:*

$$(\boldsymbol{\nabla}\wedge F)^* = \boldsymbol{\nabla}\rfloor F^*, \qquad (\boldsymbol{\nabla}\rfloor F)^* = \boldsymbol{\nabla}\wedge F^*.$$

Proof. (1) The frame is constant, so ∂_i commutes with it and $\boldsymbol{\nabla}\wedge F = e^j\wedge\partial_j F = \sum_{I,j}\partial_j\omega_I\, e^j\wedge e^{i_1}\wedge\cdots\wedge e^{i_k}$, which is the image of (M1:12.1). (2) The standard formula for the codifferential is $\delta = -\sum_i \iota_{e_i}\partial_i$ in Euclidean space, and $\iota_{e_i} = e_i\rfloor$ on multivectors; so $\delta = -\boldsymbol{\nabla}\rfloor$. Then $\boldsymbol{\nabla} = \boldsymbol{\nabla}\rfloor + \boldsymbol{\nabla}\wedge = -\delta + \mathrm{d}$. (3) $\boldsymbol{\nabla}\wedge(\boldsymbol{\nabla}\wedge F) = e^i\wedge e^j\wedge\partial_i\partial_j F = 0$ since $e^i\wedge e^j$ is antisymmetric and $\partial_i\partial_j$ symmetric (this is $\mathrm{d}^2 = 0$ again). Similarly, by Proposition 2.3(3), $\boldsymbol{\nabla}\rfloor(\boldsymbol{\nabla}\rfloor F) = (e^i\wedge e^j)\rfloor\partial_i\partial_j F = 0$. (4) $\boldsymbol{\nabla}\boldsymbol{\nabla} = e^i e^j\partial_i\partial_j = (e^i\cdot e^j)\partial_i\partial_j$ because the antisymmetric part $e^i\wedge e^j\partial_i\partial_j$ cancels. Expanding $\boldsymbol{\nabla}(\boldsymbol{\nabla}F)$ by grades and using (3) leaves the two cross terms, which equal $-(\mathrm{d}\delta+\delta\mathrm{d})$ by (1)–(2). (5) is Theorem 4.4(3) applied with $A = \boldsymbol{\nabla}$ (the identities are linear in A and the derivatives ∂_i commute with the constant dual I^{-1}). □

Remark 8.3

Parts (2) and (4) say that the Dirac operator ∇ is a "square root" of the Laplacian, $\nabla^2 = -\Delta_{\text{Hodge}}$, which splits as $\nabla = \mathrm{d} - \delta$. Item (5) shows why the codifferential is $\pm \star \mathrm{d} \star$: it is the *dual* of the exterior derivative, just as contraction is the dual of the wedge.

Example 8.4 — Vector calculus in $\mathcal{G}(\mathbb{R}^3)$

Let f be a scalar field and $\mathbf{a}$ a vector field. Then

$$\nabla f = \operatorname{grad} f, \qquad \nabla \mathbf{a} = \nabla \cdot \mathbf{a} + \nabla \wedge \mathbf{a} = \operatorname{div} \mathbf{a} + I \operatorname{curl} \mathbf{a},$$

using Proposition 4.7 ($\nabla \wedge \mathbf{a} = I(\nabla \times \mathbf{a})$). Since $\operatorname{curl} \mathbf{a} = (\nabla \wedge \mathbf{a})^*$ (Proposition 4.7), the two classical identities are both instances of $\nabla \wedge \nabla \wedge = 0$: $\operatorname{curl} \operatorname{grad} f = (\nabla \wedge \nabla f)^* = 0$, and, by (4.1), $\operatorname{div} \operatorname{curl} \mathbf{a} = \nabla \rfloor (\nabla \wedge \mathbf{a})^* = (\nabla \wedge \nabla \wedge \mathbf{a})^* = 0$. Finally $\nabla^2 = \nabla \cdot \nabla$ is the Laplacian.

Example 8.5 — Maxwell's equations in one line

In the spacetime algebra $\mathcal{G}(\mathbb{R}^{1,3})$ the electromagnetic field is the bivector $F = \mathbf{E} + I\mathbf{B}$ (with $\mathbf{E}$, $\mathbf{B}$ the electric and magnetic fields written as relative vectors, i.e. as bivectors of spacetime) and (in suitable units) the four Maxwell equations are the single equation

$$\nabla F = J, \qquad \text{i.e.} \qquad \underbrace{\nabla \wedge F = 0}_{\mathrm{d}F=0 \text{ (homogeneous)}}, \qquad \underbrace{\nabla \rfloor F = J}_{\text{inhomogeneous, source } J}.$$

The two pieces are precisely the exterior derivative and the codifferential (a dual exterior derivative) of F. The first equation has a potential, $F = \nabla \wedge A$, by the Poincaré lemma.

Intuition

In geometric algebra the derivative is *literally a vector*, $\nabla = e^i \partial_i$, so it multiplies a field F with the geometric product, and the geometric product $ab = a \cdot b + a \wedge b$ splits into a lower-grade and a higher-grade part:

$$\underbrace{\nabla F}_{\text{one derivative}} = \underbrace{\nabla \rfloor F}_{\text{divergence-like (grade}\downarrow\text{)}} + \underbrace{\nabla \wedge F}_{\text{curl-like = exterior derivative (grade}\uparrow\text{)}}.$$

The exterior derivative is thus the "wedge half" of a single, invertible derivative. Duality then converts the wedge half into the contraction half, so that curl and divergence are *the same operation seen through the dual* ($\star$): that is why Maxwell's equations, which look like two different kinds of equation in vector calculus, become one equation. Everything in this section is just $ab = a \cdot b + a \wedge b$ with a replaced by ∇.

9 Synthesis II: one calculus, three languages

Prerequisites

Part M1:I; M1:8 (The metric tensor and the musical isomorphisms); M1:9 (The exterior algebra and the wedge product); M1:12 (Tensor fields and differential forms); M1:14 (From 1-forms to k-forms: integration is full contraction); M1:15 (Stokes' theorem as it is understood today); M1:16 (The Gauss (divergence) theorem); 4 (Duality: the pseudoscalar, the dual, the cross product and the Hodge star); 5 (Synthesis I: contraction, duality and the role of the metric); 7 (The Lie derivative and Cartan's formula); 8 (The exterior derivative in geometric algebra)

Synthesis: what this section does

In $\mathbb{R}^3$ with its Euclidean metric and orientation, the 19th-century vector calculus, the calculus of differential forms and the geometric calculus are three notations for one and the same set of operations. Table 1 is the dictionary; the example and the two statements below show how to use it.

9.1 The dictionary in $\mathbb{R}^3$

In the table, $\mathrm{vol} = \mathrm{d}x \wedge \mathrm{d}y \wedge \mathrm{d}z$ is the *volume form* of $\mathbb{R}^3$ (it gives volume 1 to the unit cube), and the *flux* 2*-form* of a vector field F is $\iota_F \mathrm{vol}$, computed in (M1:9.12); Section M1:14.1 explains why its integral over a surface is the flux of F. The Hodge star on forms is the one of Section 4.

Object	Vector calculus	Forms	Geometric algebra
gradient	$\operatorname{grad} f$	$\mathrm{d}f = (\operatorname{grad} f)^\flat$	∇f
curl	$\operatorname{curl} F$	$\mathrm{d}F^\flat = \iota_{\operatorname{curl} F}\mathrm{vol} = \star(\operatorname{curl} F)^\flat$	$\nabla \wedge F = I \operatorname{curl} F$
divergence	$\operatorname{div} F$	$\mathrm{d}(\iota_F \mathrm{vol}) = (\operatorname{div} F)\mathrm{vol}$	$\nabla \rfloor F = \nabla \cdot F$
flux 2-form	$F \cdot n\, \mathrm{d}A$	$\iota_F \mathrm{vol} = \star F^\flat$	$FI = IF$
cross product	$a \times b$	$\big(\star(a^\flat \wedge b^\flat)\big)^\sharp$	$(a \wedge b)I^{-1}$
$\mathrm{dd} = 0$	$\operatorname{curl}\operatorname{grad} = 0$, $\operatorname{div}\operatorname{curl} = 0$	$\mathrm{dd} = 0$	$\nabla \wedge \nabla \wedge = 0$
all first derivatives at once	—	d and δ separately	$\nabla F = \nabla \rfloor F + \nabla \wedge F$

Table 1: The calculus of $\mathbb{R}^3$ in three languages (Euclidean metric, orientation $I = e_1e_2e_3$, $I^2 = -1$; forms and multivectors identified by $\mathrm{d}x^i \leftrightarrow e_i$).

Proposition 9.1 (three-language identities)**.** *For a vector field $F = F^i e_i$ on $\mathbb{R}^3$ (identified with $F^\flat = F_i\,\mathrm{d}x^i$):*

(i) $\iota_F \mathrm{vol} = \star F^\flat \leftrightarrow FI$, *with components* $F^1\mathrm{d}y \wedge \mathrm{d}z + F^2\mathrm{d}z \wedge \mathrm{d}x + F^3\mathrm{d}x \wedge \mathrm{d}y \leftrightarrow F^1e_2e_3 + F^2e_3e_1 + F^3e_1e_2$;

(ii) $\mathrm{d}F^\flat = \iota_{\operatorname{curl} F}\mathrm{vol} \leftrightarrow \nabla \wedge F = I \operatorname{curl} F$;

(iii) $\mathrm{d}(\iota_F \mathrm{vol}) = (\operatorname{div} F)\mathrm{vol} \leftrightarrow \nabla \wedge (FI) = (\nabla \cdot F)\, I$.

Proof. (i) By Theorem 4.10, $\star F = \widetilde{F}I = FI$ for a vector; $e_1I = e_2e_3$, $e_2I = e_3e_1$, $e_3I = e_1e_2$ (e.g. $e_2e_1e_2e_3 = -e_1e_2e_2e_3 = -e_1e_3 = e_3e_1$). Compare with the formula (M1:9.12) for $\iota_F \mathrm{vol}$ of

Section M1:9.6. (ii) $\nabla \wedge F = \sum_{i<j}(\partial_i F_j - \partial_j F_i)\, e_i e_j = (\partial_2 F_3 - \partial_3 F_2)e_2e_3 + (\partial_3 F_1 - \partial_1 F_3)e_3e_1 + (\partial_1 F_2 - \partial_2 F_1)e_1e_2$, which is $I \operatorname{curl} F$ by the same table of $e_i I$; the forms side is the computation of $\mathrm{d}F^\flat$ in Section M1:12.3. □

Example 9.2 — all three languages on one field

Let $F = (-y, x, z)$ on $\mathbb{R}^3$. *Vector calculus:* $\operatorname{curl} F = (0,0,2)$, $\operatorname{div} F = 1$. *Forms:* $F^\flat = -y\,\mathrm{d}x + x\,\mathrm{d}y + z\,\mathrm{d}z$, so $\mathrm{d}F^\flat = -\mathrm{d}y \wedge \mathrm{d}x + \mathrm{d}x \wedge \mathrm{d}y = 2\,\mathrm{d}x \wedge \mathrm{d}y = \iota_{(0,0,2)}\mathrm{vol}$ ✓; and $\iota_F \mathrm{vol} = -y\,\mathrm{d}y \wedge \mathrm{d}z + x\,\mathrm{d}z \wedge \mathrm{d}x + z\,\mathrm{d}x \wedge \mathrm{d}y$, whose d is $\partial_z z\,\mathrm{d}z \wedge \mathrm{d}x \wedge \mathrm{d}y = \mathrm{vol}$ (the other two terms contribute $\partial_x(-y) = \partial_y(x) = 0$), i.e. $\operatorname{div} F = 1$ ✓. *Geometric algebra:* the single product $\nabla F = \nabla \rfloor F + \nabla \wedge F = 1 + I(0,0,2) = 1 + 2\,e_1e_2$ contains both answers at once: scalar part $\operatorname{div} F = 1$, bivector part $I \operatorname{curl} F = 2e_1e_2$.

9.2 One reason for three identities

Synthesis: $\mathrm{dd} = 0$, "curl grad $= 0$" and "div curl $= 0$"

In forms, $\mathrm{dd} = 0$ because mixed partial derivatives commute while $\mathrm{d}x^j \wedge \mathrm{d}x^i$ is antisymmetric (Example M1:12.7). In geometric algebra, $\nabla \wedge \nabla \wedge A = 0$ for the same reason: $\nabla \wedge \nabla = e^i_\sharp \wedge e^j_\sharp\, \partial_i \partial_j = 0$. Translated back to vector calculus, the dd on 0-forms is $\operatorname{curl}\operatorname{grad} = 0$ and on 1-forms is $\operatorname{div}\operatorname{curl} = 0$ (use $\mathrm{dd} = 0$ together with the dictionary above). All three are one fact: *the second derivative of a smooth function is symmetric*. The same fact is what fails for covariant derivatives, whose antisymmetrised second derivative is the curvature (Section 14).

9.3 Maxwell's equations in three languages

Language	Equations	Needs a metric?
Vector calculus	$\operatorname{curl} E = -\partial_t B$, $\operatorname{div} B = 0$, $\operatorname{div} D = \rho$, $\operatorname{curl} H = j + \partial_t D$	curl and div do
Forms (Section GR:3.1)	$\mathrm{d}F = 0$, $\mathrm{d} \star F = J$	only $\star$ (and only its conformal class)
Geometric algebra (spacetime algebra)	$\nabla F = J$ with $F = E + IB$	the geometric product does; both equations are one

The first form equation $\mathrm{d}F = 0$ (the homogeneous equations) is metric-free and is the "wedge half" $\nabla \wedge F = 0$; the second ($\mathrm{d} \star F = J$) is the "contraction half" $\nabla \rfloor F = J$, equivalent by duality (Section 4). The sign conventions differ with the signature convention (Section 5.3).

Key points of Part II

- Manifold = patchwork of charts; tangent and cotangent spaces carry all the algebra of Part M1:I pointwise; forms are the antisymmetric covariant tensor fields.
- d is the unique graded-Leibniz extension of the differential of functions with $\mathrm{dd} = 0$; $\mathcal{L}_X = \mathrm{d}\iota_X + \iota_X \mathrm{d}$.
- In $\mathbb{R}^3$ grad, curl, div are d in degrees $0, 1, 2$, dressed with $\flat, \sharp, \star$; geometric algebra packages them as one operator $\nabla = \nabla\cdot + \nabla\wedge$.

9.4 Classical corollaries

Let $\mathbb{R}^3$ carry its Euclidean metric g, orientation and volume form $\mathrm{vol} = \mathrm{d}x \wedge \mathrm{d}y \wedge \mathrm{d}z$. With the musical isomorphisms $\flat, \sharp$ and the Hodge star $\star$ one has the dictionary (the one of Section 9, written with $\star$)

$$\operatorname{grad} f = (\mathrm{d}f)^{\sharp}, \qquad \operatorname{curl} F = \big(\star\, \mathrm{d} F^{\flat}\big)^{\sharp}, \qquad \operatorname{div} F = \star\, \mathrm{d} \star F^{\flat}, \qquad \iota_F \mathrm{vol} = \star F^{\flat}.$$

Applying Theorem M1:15.1 to forms of degree $0, 1, 2$ yields:

Form of Stokes	**Differential forms**	**Classical statement**
$n = 1$	$\int_a^b \mathrm{d}f = f(b) - f(a)$	fundamental theorem of calculus / gradient theorem
$n = 2$ (planar D)	$\int_D \mathrm{d}\omega = \oint_{\partial D} \omega$	Green: $\oint P\,\mathrm{d}x + Q\,\mathrm{d}y = \iint (Q_x - P_y)\,\mathrm{d}A$
$n = 2$ (surface $S \subset \mathbb{R}^3$)	$\int_S \mathrm{d}F^{\flat} = \oint_{\partial S} F^{\flat}$	Kelvin–Stokes: $\oint F \cdot \mathrm{d}r = \iint (\operatorname{curl} F) \cdot n\,\mathrm{d}A$
$n = 3$	$\int_\Omega \mathrm{d}(\iota_F \mathrm{vol}) = \int_{\partial\Omega} \iota_F \mathrm{vol}$	Gauss: $\iiint \operatorname{div} F\,\mathrm{d}V = \iint_{\partial\Omega} F \cdot n\,\mathrm{d}A$

Verification of the dictionary, step by step

(i) Curl and *(ii) divergence* were verified in Section M1:12.3: $\mathrm{d}F^{\flat} = \iota_{\operatorname{curl} F}\mathrm{vol}$ and $\mathrm{d}(\iota_F \mathrm{vol}) = (\operatorname{div} F)\,\mathrm{vol}$.

(iii) Kelvin–Stokes. For an oriented surface $S \subset \mathbb{R}^3$ with unit normal n the restriction satisfies $\iota_G \mathrm{vol}|_S = \langle G, n\rangle\,\mathrm{d}A$ (same computation as in Lemma M1:16.1, with $e_1 = n$ and e_2, e_3 tangent to S). Then

$$\oint_{\partial S} F \cdot \mathrm{d}r = \oint_{\partial S} F^{\flat} \overset{\text{Stokes}}{=} \int_S \mathrm{d}F^{\flat} = \int_S \iota_{\operatorname{curl} F}\mathrm{vol} = \iint_S \langle \operatorname{curl} F, n\rangle\,\mathrm{d}A.$$

(iv) Gauss. $\int_\Omega (\operatorname{div} F)\,\mathrm{d}V \overset{\text{(ii)}}{=} \int_\Omega \mathrm{d}(\iota_F \mathrm{vol}) \overset{\text{Stokes}}{=} \int_{\partial\Omega} \iota_F \mathrm{vol} = \iint_{\partial\Omega} \langle F, n\rangle\,\mathrm{d}A$, again by the restriction formula.

Part III Connections, geodesics and curvature

Part M1:III needed only a smooth structure and an orientation. To differentiate *vectors* we must compare tangent spaces at nearby points, and that requires extra structure: a *connection*. A metric singles out one canonical connection (Levi-Civita), whose geodesics are the straightest curves, and whose failure to be "flat" is the Riemann curvature tensor: the mathematical content of gravity in general relativity.

10 Connections and the covariant derivative

Prerequisites

M1:7 (Multilinear maps, tensors and the tensor algebra); M1:11 (Tangent and cotangent spaces); M1:12 (Tensor fields and differential forms)

10.1 Definition

Intuition: derivative = "compare with the parallel copy"

We choose a rule that says what it means for a vector to be "kept constant" while we move along a curve: this is **parallel transport** Π. Then the derivative of a vector field Y along X is the limit of $\frac{1}{\varepsilon}\big(\Pi^{-1}(Y_{\phi_\varepsilon(p)}) - Y_p\big)$: how much Y differs from the parallel copy of itself. Infinitesimally, the same information is encoded algebraically by a *connection* ∇. This is the analogue of the relation between a vector field X (infinitesimal) and its flow ϕ (finite).

Formal idea: connection (affine connection)

A *connection* on M is a map $\mathfrak{X}(M) \times \mathfrak{X}(M) \to \mathfrak{X}(M)$, $(X, Y) \mapsto \nabla_X Y$ such that

(i) it is $\mathbb{R}$-bilinear;
(ii) $\nabla_{fX} Y = f\,\nabla_X Y$ for $f \in C^\infty(M)$ (C^∞-linear in the direction X: only X_p matters);
(iii) $\nabla_X(fY) = X(f)\,Y + f\,\nabla_X Y$ (Leibniz/product rule in Y).

In a chart the *Christoffel symbols* $\Gamma^k{}_{ij}(x)$ are defined by

$$\nabla_{\partial_i}\partial_j = \Gamma^k{}_{ij}\,\partial_k,$$

and then for $X = X^i\partial_i$, $Y = Y^j\partial_j$:

$$\nabla_X Y = X^i\Big(\frac{\partial Y^k}{\partial x^i} + \Gamma^k{}_{ij}Y^j\Big)\partial_k. \tag{10.1}$$

Given $\Gamma^k{}_{ij}$ in one chart, ∇ is completely determined there. Conversely any choice of smooth functions $\Gamma^k{}_{ij}$ defines a connection on the chart domain.

Intuition: Christoffel symbols = correction terms

$\partial_i Y^k$ is the change of the *components* of Y; $\Gamma^k{}_{ij} Y^j$ is the correction due to the fact that the *basis vectors* themselves change from point to point (if you walk along a circle, the basis $\partial_r, \partial_\varphi$ rotates). Even a constant vector field $Y = \text{const}$ in the plane has $\partial_i Y^k \neq 0$ in polar coordinates; the Γ term compensates exactly.

Formal idea: Γ is not a tensor

Under a coordinate change $x \mapsto \tilde{x}$,

$$\tilde{\Gamma}^{\tilde{k}}{}_{\tilde{i}\tilde{j}} = \frac{\partial x^i}{\partial \tilde{x}^{\tilde{i}}} \frac{\partial x^j}{\partial \tilde{x}^{\tilde{j}}} \Gamma^k{}_{ij} \frac{\partial \tilde{x}^{\tilde{k}}}{\partial x^k} + \frac{\partial^2 x^k}{\partial \tilde{x}^{\tilde{i}} \partial \tilde{x}^{\tilde{j}}} \frac{\partial \tilde{x}^{\tilde{k}}}{\partial x^k}. \tag{10.2}$$

The first term is the tensor rule (M1:7.1); the second is the extra term that spoils tensoriality. (Consequence: Γ can vanish in one chart and not in another, as below.) The two connections ∇^0, ∇^1 differ by a genuine tensor, so connections form an affine space. (Derivation of (10.2): Identity C.35 in the Glossary.)

Example 10.1 — Christoffel symbols of the Euclidean connection in polar coordinates

Let ∇^0 be the connection with $\Gamma = 0$ in Cartesian coordinates (x, y) (*i.e.* $\nabla^0_X \partial_x = \nabla^0_X \partial_y = 0$). In the formula (10.2) only the second term survives: $\tilde{\Gamma}^{\tilde{k}}{}_{\tilde{i}\tilde{j}} = \frac{\partial^2 x^l}{\partial \tilde{x}^{\tilde{i}} \partial \tilde{x}^{\tilde{j}}} \frac{\partial \tilde{x}^{\tilde{k}}}{\partial x^l}$, with $x = r\cos\varphi$, $y = r\sin\varphi$, $\partial_x r = \cos\varphi$, $\partial_y r = \sin\varphi$, $\partial_x \varphi = -\sin\varphi/r$, $\partial_y \varphi = \cos\varphi/r$:

$$\Gamma^r{}_{\varphi\varphi} = (-r\cos\varphi)(\cos\varphi) + (-r\sin\varphi)(\sin\varphi) = -r,$$
$$\Gamma^\varphi{}_{r\varphi} = \Gamma^\varphi{}_{\varphi r} = (-\sin\varphi)\left(\tfrac{-\sin\varphi}{r}\right) + (\cos\varphi)\left(\tfrac{\cos\varphi}{r}\right) = \tfrac{1}{r},$$

and all the others vanish ($\Gamma^r{}_{rr} = \Gamma^r{}_{r\varphi} = \Gamma^\varphi{}_{rr} = \Gamma^\varphi{}_{\varphi\varphi} = 0$). ✓ Interpretation: $\nabla_{\partial_\varphi} \partial_\varphi = -r\,\partial_r$ (the basis vector ∂_φ turns inwards as you go around) and $\nabla_{\partial_r} \partial_\varphi = \frac{1}{r} \partial_\varphi$ (∂_φ grows linearly with r).

Example 10.2 — a connection is extra structure: the "cylindrical" connection

On $\mathbb{R}^2 \setminus \{0\}$ define ∇^1 by declaring $\Gamma = 0$ in the *polar* chart, *i.e.* $\nabla^1_{\partial_a} \partial_b = 0$ for $a, b \in \{r, \varphi\}$. This is a perfectly good connection, but **different** from ∇^0 (since $\Gamma^r{}_{\varphi\varphi} = -r \neq 0$ for ∇^0). Differences: ∇^1 keeps $\partial_r, \partial_\varphi$ "constant" (so arrows rotate with the polar frame) while ∇^0 keeps ∂_x, ∂_y constant. Their geodesics differ: for ∇^1 geodesics satisfy $\ddot{r} = \ddot{\varphi} = 0$ (spirals $r = r_0 + at$, $\varphi = \varphi_0 + bt$), for ∇^0 they are straight lines.

10.2 Covariant derivative of covectors and general tensors

Formal idea: extension to all tensors

There is a unique extension of ∇_X to $\Pi^\ell_k(M)$ (still of the same type) such that

- $\nabla_X f = X(f)$ for functions;
- $\nabla_X(\tau_1 \otimes \tau_2) = \nabla_X \tau_1 \otimes \tau_2 + \tau_1 \otimes \nabla_X \tau_2$;
- it commutes with contractions, equivalently $\nabla_X\big(\tau(v_1, \dots, v_k, \alpha_1, \dots, \alpha_\ell)\big) = (\nabla_X \tau)(\dots) + \sum_i \tau(\dots, \nabla_X v_i, \dots) + \sum_j \tau(\dots, \nabla_X \alpha_j, \dots)$.

For a 1-form $(\nabla_X \omega)(Y) = X(\omega(Y)) - \omega(\nabla_X Y)$. In coordinates (*semicolon / index notation*, with the derivative index first):

$$\begin{aligned} \nabla_i V^k &= \partial_i V^k + \Gamma^k{}_{ij} V^j, \\ \nabla_i \omega_j &= \partial_i \omega_j - \Gamma^k{}_{ij} \omega_k, \\ \nabla_r T^a{}_b &= \partial_r T^a{}_b + \Gamma^a{}_{rs} T^s{}_b - \Gamma^s{}_{rb} T^a{}_s. \end{aligned}$$

Rule of thumb: every upper index gets a $+\Gamma$ term and every lower index a $-\Gamma$ term. The result $\nabla\tau$ is a tensor of type $(\ell, k+1)$ with components $\tau^{\cdots}{}_{\ldots;r} = \nabla_r \tau^{\cdots}{}_{\ldots}$ even though $\partial_r \tau$ and Γ are not tensors separately: the non-tensorial parts cancel.

Intuition: covectors are the "dual" behaviour

Since $\omega(Y)$ must obey the ordinary product rule for a scalar, ∇ on covectors *must* compensate the connection acting on vectors; this forces the minus sign.

Example 10.3 — divergence in polar coordinates

For a vector field V, $\operatorname{div} V := \nabla_i V^i = \partial_i V^i + \Gamma^i{}_{ij} V^j$. With the polar Γ: $\Gamma^i{}_{ir} = \Gamma^r{}_{rr} + \Gamma^\varphi{}_{\varphi r} = \frac{1}{r}$ and $\Gamma^i{}_{i\varphi} = 0$, therefore

$$\operatorname{div} V = \partial_r V^r + \partial_\varphi V^\varphi + \frac{V^r}{r} = \frac{1}{r}\partial_r (r V^r) + \partial_\varphi V^\varphi,$$

the familiar formula (with V^φ the component on ∂_φ). No ad-hoc "scale factors" were needed: they are all contained in Γ.

10.3 Parallel transport

Formal idea: parallel transport along a curve

For a smooth curve $\gamma(t)$ and a vector field Y along γ (a smooth assignment $t \mapsto Y_{\gamma(t)} \in T_{\gamma(t)}M$), define $\nabla_{\dot\gamma} Y := \nabla_{\hat X} \hat Y$ for any extensions $\hat X, \hat Y$ of $\dot\gamma, Y$ to fields on M; the result does not depend on the extensions. In coordinates $\gamma = (x^i(t))$:

$$(\nabla_{\dot\gamma} Y)^k = \frac{\mathrm{d}Y^k}{\mathrm{d}t} + \Gamma^k{}_{ij}(\gamma(t))\, \dot x^i\, Y^j.$$

Theorem. Given $v_0 \in T_{\gamma(t_0)}M$ there is a unique field Y along γ with $Y_{\gamma(t_0)} = v_0$ and $\nabla_{\dot\gamma} Y = 0$. (In coordinates this is the linear ODE

$$\frac{\mathrm{d}Y^k}{\mathrm{d}t} + \Gamma^k{}_{ij}(\gamma(t))\, \dot x^i\, Y^j = 0, \tag{10.3}$$

so existence and uniqueness follow from ODE theory; a general curve is split into finitely many chart pieces.) The *parallel transport* is the linear isomorphism $\Pi^{\gamma}_{\gamma(t_0),\gamma(t_1)} : T_{\gamma(t_0)}M \to T_{\gamma(t_1)}M$, $v_0 \mapsto Y_{\gamma(t_1)}$.

Intuition: carrying an arrow without turning it

Like carrying a spear while walking and trying not to rotate it "relative to the ground". On a curved surface this is ambiguous globally, but infinitesimally it is perfectly well defined by (10.3). In a flat space with ∇^0 and Cartesian coordinates ($\Gamma = 0$) parallel transport leaves components unchanged ($\Pi(v_0) = v_0$ via $T_p\mathbb{R}^m \cong \mathbb{R}^m$).

On a curved surface the result of parallel transport depends on the path; the sphere is worked out in Example 11.2 of Section 11, once its Christoffel symbols are available.

10.4 Geodesics

Formal idea: geodesic

A curve $\gamma(t)$ is a *geodesic* of ∇ if $\nabla_{\dot\gamma}\dot\gamma = 0$ for all t (it parallel-transports its own tangent vector: "auto-parallelism"). In coordinates:

$$\ddot{x}^k + \Gamma^k{}_{ij}(x(t))\,\dot{x}^i\dot{x}^j = 0. \tag{10.4}$$

This is a second-order ODE: given a point and an initial velocity there is a unique maximal geodesic. If one reparametrizes, $\gamma(t) = \rho(s(t))$, then γ is a geodesic as a set iff ρ satisfies $\frac{\mathrm{d}^2y^k}{\mathrm{d}s^2} + \Gamma^k{}_{ij}\frac{\mathrm{d}y^i}{\mathrm{d}s}\frac{\mathrm{d}y^j}{\mathrm{d}s} = f(s)\frac{\mathrm{d}y^k}{\mathrm{d}s}$ with $f = -s''/(s')^2$; geodesics with $f = 0$ are called *affinely parametrized* (and (10.4) is that case).

Intuition: the straightest possible path; free fall

A geodesic is a curve with zero covariant acceleration: "no one pushes me, I just follow the geometry". In $\mathbb{R}^m$ with ∇^0 it is $\ddot{x} = 0$: uniform straight motion. The term $-\Gamma^k{}_{ij}\dot{x}^i\dot{x}^j$ is the acceleration that *coordinates* see; in GR it is precisely what we call the "gravitational force" (and in polar coordinates, the "centrifugal" force). In GR: **freely falling particles move along timelike geodesics; light along null geodesics.**

Example 10.4 — straight lines in polar coordinates ⇒ centrifugal force

With $\Gamma^r{}_{\varphi\varphi} = -r$, $\Gamma^\varphi{}_{r\varphi} = 1/r$, equation (10.4) reads

$$\ddot{r} - r\dot{\varphi}^2 = 0, \qquad \ddot{\varphi} + \frac{2}{r}\dot{r}\dot{\varphi} = 0.$$

The second equation gives $\frac{\mathrm{d}}{\mathrm{d}t}(r^2\dot\varphi) = r(r\ddot\varphi + 2\dot r\dot\varphi) = 0$: $L = r^2\dot\varphi$ is conserved (angular momentum). Then $\ddot r = L^2/r^3$: a repulsive "centrifugal" term appears even though nothing is pushing the particle; it is just a straight line seen in a rotating-frame-like coordinate system. *Check with the line* $x = d$, $y = vt$: $r^2 = d^2 + v^2t^2$, $\varphi = \arctan(vt/d)$, so $r^2\dot\varphi = vd$ is constant, and $\ddot r = v^2d^2/r^3 = L^2/r^3$ with $L = vd$. ✓

The geodesics of the sphere (great circles) are found in Example 11.3 of Section 11.

Exercises

1. Verify from (10.2) that $\tilde\Gamma^{\tilde k}{}_{\tilde i\tilde j}$ is symmetric in its lower indices if Γ is.
2. Write $\nabla_X(f\omega) = X(f)\omega + f\nabla_X\omega$ and check it in coordinates.
3. Show that the geodesic equation implies that $g(\dot\gamma,\dot\gamma)$ is constant when ∇ is the Levi-

Civita connection (next section).

11 Torsion, the Levi-Civita connection and geodesics

Prerequisites

M1:8 (The metric tensor and the musical isomorphisms); 10 (Connections and the covariant derivative)

Formal idea: torsion

The *torsion* of ∇ is $\mathrm{Tor}(X,Y) = \nabla_X Y - \nabla_Y X - [X,Y]$. It is $C^\infty(M)$-linear in both arguments, hence given by a tensor $T \in \Pi^1_2(M)$; in a chart $T^i{}_{jk} = \Gamma^i{}_{jk} - \Gamma^i{}_{kj}$. It is antisymmetric: $T(X,Y) = -T(Y,X)$. The connection is called *symmetric* (or *torsion-free*) if $T = 0$, which in coordinates means $\Gamma^i{}_{jk} = \Gamma^i{}_{kj}$.

Intuition: torsion = twisting of the frames

Torsion measures whether an infinitesimal parallelogram built by parallel-transporting two small vectors along each other closes up. Torsion-free means "no screw-like twisting". In GR we always assume $T = 0$.

Formal idea: metric compatibility and the Levi-Civita theorem

Let g be a metric. A connection is *metric* if $\nabla_X g = 0$ for all X, *i.e.* $X\big(g(Y,Z)\big) = g(\nabla_X Y, Z) + g(Y, \nabla_X Z)$.

Theorem (Fundamental theorem of (pseudo-)Riemannian geometry). For every metric g there is a *unique* connection $\nabla = \nabla^{LC}$ that is (i) metric and (ii) torsion-free. It is determined by the *Koszul formula*

$$\begin{aligned} 2g(\nabla_X Y, Z) = {} & X\big(g(Y,Z)\big) + Y\big(g(X,Z)\big) - Z\big(g(X,Y)\big) \\ & + g([Z,X],Y) + g([X,Y],Z) - g(X,[Y,Z]). \end{aligned}$$

Idea of the proof. Write the metric property for $(X;Y,Z)$, $(Y;Z,X)$, $(Z;X,Y)$, add the first two and subtract the third, and use $\nabla_X Y - \nabla_Y X = [X,Y]$ to eliminate all covariant derivatives except $g(\nabla_X Y, Z)$. Non-degeneracy of g then determines $\nabla_X Y$.

In coordinates, taking $X = \partial_i, Y = \partial_j, Z = \partial_l$ (all brackets vanish):

$$\boxed{\Gamma^k{}_{ij} = \tfrac{1}{2} g^{kl}\big(\partial_i g_{jl} + \partial_j g_{il} - \partial_l g_{ij}\big)} \tag{11.1}$$

with (g^{kl}) the inverse matrix of (g_{ij}).

Intuition: the "natural" connection of a metric

The Levi-Civita connection is the unique way of doing parallel transport such that (i) **lengths and angles are preserved**, since $g(\Pi v, \Pi w) = g(v,w)$ and (ii) there is **no torsion**. It is therefore a function of the metric alone, $g \mapsto \nabla^{LC}$: every construction from ∇^{LC} is a

construction from g. Geodesics of ∇^{LC} are called "geodesics of g". Also, ∇ commutes with raising and lowering of indices.

Example 11.1 — Christoffel symbols from the metric

Polar plane: $g_{rr} = 1$, $g_{\varphi\varphi} = r^2$. The only non-zero derivative is $\partial_r g_{\varphi\varphi} = 2r$. From (11.1): $\Gamma^r{}_{\varphi\varphi} = \frac{1}{2}g^{rr}(-\partial_r g_{\varphi\varphi}) = -r$, $\Gamma^\varphi{}_{r\varphi} = \frac{1}{2}g^{\varphi\varphi}\partial_r g_{\varphi\varphi} = \frac{1}{r}$. ✓ (The same as before: the Euclidean connection *is* the Levi-Civita connection of the Euclidean metric.)

Unit sphere: $g_{\theta\theta} = 1$, $g_{\varphi\varphi} = \sin^2\theta$, $\partial_\theta g_{\varphi\varphi} = 2\sin\theta\cos\theta$:

$$\Gamma^\theta{}_{\varphi\varphi} = -\tfrac{1}{2}\partial_\theta g_{\varphi\varphi} = -\sin\theta\cos\theta, \qquad \Gamma^\varphi{}_{\theta\varphi} = \Gamma^\varphi{}_{\varphi\theta} = \tfrac{1}{2}\frac{1}{\sin^2\theta}\,\partial_\theta g_{\varphi\varphi} = \cot\theta,$$

all others zero. Radius a gives the same symbols (a constant rescaling of g does not change Γ).

Expanding universe (1+1 dimensions): $g = -\mathrm{d}t\otimes\mathrm{d}t + a(t)^2\,\mathrm{d}x\otimes\mathrm{d}x$. Then $\Gamma^t{}_{xx} = \frac{1}{2}g^{tt}(-\partial_t g_{xx}) = \frac{1}{2}(-1)(-2a\dot{a}) = a\dot{a}$, $\Gamma^x{}_{tx} = \frac{1}{2}g^{xx}\partial_t g_{xx} = \dot{a}/a$.

Example 11.2 — parallel transport on the sphere: path-dependence

For the unit sphere (metric $\mathrm{d}\theta^2 + \sin^2\theta\,\mathrm{d}\varphi^2$; Christoffel symbols of the previous example): $\Gamma^\theta{}_{\varphi\varphi} = -\sin\theta\cos\theta$, $\Gamma^\varphi{}_{\theta\varphi} = \cot\theta$. *Along the equator* $\theta = \pi/2$, $\dot\theta = 0$, $\dot\varphi = 1$: both symbols vanish at $\theta = \pi/2$, so (10.3) gives $\dot{Y}^\theta = \dot{Y}^\varphi = 0$: the components (Y^θ, Y^φ) stay constant. *Along a meridian* ($\varphi = $ const, $\dot\theta = 1$): $\dot{Y}^\theta = 0$ and $\dot{Y}^\varphi + \cot\theta\, Y^\varphi = 0$, so Y^θ is constant, $Y^\varphi \propto 1/\sin\theta$ (hence its length $\sin\theta\,|Y^\varphi|$ is constant). *A triangle.* Go from the north pole down the meridian $\varphi = 0$ to the equator, along the equator to $\varphi = \pi/2$, and back up to the pole. Start at the pole with a vector pointing along the first meridian's direction "away from the pole" (θ-direction at $\varphi = 0$). Transported down the first meridian it remains tangent to it; along the equator it stays pointing towards the south (constant Y^θ), perpendicular to the path; going back up the second meridian it remains the same way, now pointing *along* the second meridian, away from the pole. At the end the vector has rotated by $\pi/2$ with respect to the initial one. The rotation angle equals the enclosed area times the curvature ($K = 1$): the octant has area $4\pi/8 = \pi/2$. **Parallel transport depends on the path precisely because the space is curved** (Section 12).

Example 11.3 — geodesic equations on the unit sphere

Using $\Gamma^\theta{}_{\varphi\varphi} = -\sin\theta\cos\theta$, $\Gamma^\varphi{}_{\theta\varphi} = \cot\theta$ in (10.4):

$$\ddot\theta - \sin\theta\cos\theta\,\dot\varphi^2 = 0, \qquad \ddot\varphi + 2\cot\theta\,\dot\theta\dot\varphi = 0.$$

Equator $\theta = \pi/2$, $\dot\theta = 0$: both equations hold for $\varphi = \omega t \Rightarrow$ the equator is a geodesic (a great circle). *Circle of latitude* $\theta = \theta_0 \neq \pi/2$, $\varphi = \omega t$: the first equation gives $\ddot\theta = \sin\theta_0\cos\theta_0\,\omega^2 \neq 0$, contradicting $\ddot\theta = 0 \Rightarrow$ it is **not** a geodesic: you must push to stay on it. All geodesics of the sphere are great circles.

11.1 Geodesics as extremals of length (variational principle)

Formal idea: geodesics extremize the energy functional

For a curve $\gamma : [t_0, t_1] \to M$ consider

$$S_1[\gamma] = \int_{t_0}^{t_1} \sqrt{g(\dot\gamma, \dot\gamma)}\, \mathrm{d}t \quad \text{(arc length)}, \qquad S_2[\gamma] = \int_{t_0}^{t_1} g(\dot\gamma, \dot\gamma)\, \mathrm{d}t \quad \text{(“energy”)}.$$

γ is a critical point of S_1 iff (after a reparametrization) it is a critical point of S_2; and γ **is a critical point of S_2 with fixed endpoints iff it is a geodesic of g**. Proof via the Euler–Lagrange equations for $\mathcal{L} = \frac{1}{2} g_{ij}(x)\dot x^i \dot x^j$:

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial \mathcal{L}}{\partial \dot x^l} - \frac{\partial \mathcal{L}}{\partial x^l} = 0 \iff g_{lj}\ddot x^j + \partial_k g_{lj}\dot x^k \dot x^j - \tfrac{1}{2}\partial_l g_{ij}\dot x^i \dot x^j = 0,$$

and multiplying by g^{ml} and symmetrizing the middle term in (k, j) gives exactly $\ddot x^m + \Gamma^m{}_{ij}\dot x^i \dot x^j = 0$ with Γ from (11.1).

Intuition: shortest path (Riemannian), longest proper time (Lorentzian)

On a sphere the shortest path between two points is an arc of a great circle (not unique for antipodal points). In Minkowski space, a straight timelike worldline *maximizes* the proper time between two events (twin paradox). The variational point of view is also the starting point of the Lagrangian formulation of GR and the reason why geodesic motion is a “principle of least action”.

Example 11.4 — Euler–Lagrange in polar coordinates

$\mathcal{L} = \frac{1}{2}(\dot r^2 + r^2\dot\varphi^2)$. $\frac{\mathrm{d}}{\mathrm{d}t}\partial_{\dot r}\mathcal{L} = \ddot r$, $\partial_r \mathcal{L} = r\dot\varphi^2$, hence $\ddot r = r\dot\varphi^2$; and $\frac{\mathrm{d}}{\mathrm{d}t}(r^2\dot\varphi) = 0$. These are exactly the geodesic equations of Example 10.4, obtained without computing a single Christoffel symbol. This is often the fastest way to find Γ: read them off as the coefficients of $-\dot x^i \dot x^j$ in the E–L equations.

Formal idea: conserved quantities

If ξ is a Killing field of g and γ is a geodesic then $g(\xi, \dot\gamma)$ is constant along γ. (Proof: $\frac{\mathrm{d}}{\mathrm{d}t} g(\xi, \dot\gamma) = g(\nabla_{\dot\gamma}\xi, \dot\gamma) + g(\xi, \nabla_{\dot\gamma}\dot\gamma) = \nabla_{(i}\xi_{j)}\dot\gamma^i\dot\gamma^j + 0 = 0$, because $\mathcal{L}_\xi g_{ij} = \nabla_i\xi_j + \nabla_j\xi_i = 0$.) The identity $\mathcal{L}_\xi g_{ij} = \nabla_i\xi_j + \nabla_j\xi_i$ for the Levi-Civita connection is proved in Proposition 14.1(b).

Example 11.5 — conservation of angular momentum

$\xi = \partial_\varphi$ is Killing for $g = \mathrm{d}r^2 + r^2\mathrm{d}\varphi^2$. Then $g(\xi, \dot\gamma) = r^2\dot\varphi$ is conserved, as found in Example 10.4. For Schwarzschild one gets the energy $E = -g(\partial_t, \dot\gamma)$ and $L = g(\partial_\varphi, \dot\gamma)$.

Example 11.6 — geodesics of an expanding universe: cosmological redshift (1+1 model)

From $\Gamma^t{}_{xx} = a\dot a$, $\Gamma^x{}_{tx} = \dot a/a$ (previous example), with affine parameter λ:

$$\ddot t + a\dot a\,\dot x^2 = 0, \qquad \ddot x + \frac{2\dot a}{a}\dot t\dot x = 0.$$

The second equation is $\frac{\mathrm{d}}{\mathrm{d}\lambda}(a^2\dot{x}) = 0$ (the translation ∂_x is Killing): $p := a^2\dot{x}$ is conserved. The *physical* momentum is the norm of the spatial velocity, $a\dot{x} = p/a(t)$: as the universe expands (a grows), the physical momentum of a free particle decreases, *i.e.* photons are **redshifted**. Light rays satisfy $g(\dot{\gamma},\dot{\gamma}) = 0 \Rightarrow dx/dt = \pm 1/a(t)$.

Exercises

1. Compute the Christoffel symbols of $\mathrm{d}s^2 = \mathrm{d}r^2 + r^2(\mathrm{d}\theta^2 + \sin^2\theta\,\mathrm{d}\varphi^2)$ (flat $\mathbb{R}^3$ in spherical coordinates).
2. Derive the geodesic equations for $\mathrm{d}s^2 = -f(r)\mathrm{d}t^2 + \mathrm{d}r^2/f(r) + r^2\mathrm{d}\Omega^2$ using E–L, and identify the conserved quantities.
3. Show that $\nabla g = 0$ implies that Π^γ preserves g (hint: $\frac{\mathrm{d}}{\mathrm{d}t}g(Y_1,Y_2)$ for parallel Y_1, Y_2).

12 The Riemann curvature tensor

Prerequisites

10 (Connections and the covariant derivative); 11 (Torsion, the Levi-Civita connection and geodesics)

Intuition: curvature = failure of parallel transport to be path independent

On a flat plane, if you parallel-transport a vector from p to q along two different paths you obtain the same vector. On a sphere you do not (Example 11.2: the octant triangle rotates the vector by $\pi/2$). Curvature measures this effect *infinitesimally*: carry v around a tiny parallelogram with sides εX and δY; it comes back changed by

$$\Delta v \;\approx\; \varepsilon\,\delta\; R(X,Y)v$$

(up to a sign depending on the orientation of the loop). Algebraically, curvature is the failure of covariant derivatives to commute: $\nabla_X\nabla_Y - \nabla_Y\nabla_X \neq \nabla_{[X,Y]}$. In GR, "gravity" is precisely curvature, the part that cannot be removed by changing coordinates.

Formal idea: Riemann curvature

For any connection ∇ and fields X, Y, Z define

$$\boxed{R(X,Y)Z = \nabla_X\nabla_Y Z - \nabla_Y\nabla_X Z - \nabla_{[X,Y]}Z}$$

This expression is $C^\infty(M)$-linear in each of X, Y, Z, so it defines a tensor $R \in \Pi^1_3(M)$, the *Riemann curvature tensor*. In a chart, $R = R^l{}_{ijk}\,\mathrm{d}x^i \otimes \mathrm{d}x^j \otimes \mathrm{d}x^k \otimes \partial_l$, i.e. $R(\partial_i,\partial_j)\partial_k = R^l{}_{ijk}\partial_l$, with

$$\boxed{R^l{}_{ijk} = \partial_i\Gamma^l{}_{jk} - \partial_j\Gamma^l{}_{ik} + \Gamma^s{}_{jk}\Gamma^l{}_{is} - \Gamma^s{}_{ik}\Gamma^l{}_{js}} \tag{12.1}$$

(derive it by expanding $\nabla_{\partial_i}\nabla_{\partial_j}\partial_k - \nabla_{\partial_j}\nabla_{\partial_i}\partial_k$; this is done in Identity C.36 of the Glossary).

Careful: index conventions

Authors place the indices differently. In Carroll's book [21] ${R^{\rho}}_{\sigma\mu\nu}$ with ρ the output, σ the input vector and $(\mu\nu)$ the pair of directions: it equals ${R^{l}}_{ijk}$ used here with $(l,k,i,j) = (\rho,\sigma,\mu,\nu)$. The sign of the whole tensor can also differ (some lecture notes use the opposite overall sign). Always check the definition before comparing formulas.

12.1 General properties

- $R(X,Y,Z) = -R(Y,X,Z)$ (antisymmetric in the first two slots).
- **First Bianchi identity.** $R(X,Y,Z) + R(Y,Z,X) + R(Z,X,Y) = \nabla_X(T(Y,Z)) + \cdots + T(X,[Y,Z]) + \ldots$ which is 0 when ∇ is torsion-free. It follows from the Jacobi identity $[X,[Y,Z]] + [Y,[Z,X]] + [Z,[X,Y]] = 0$ (torsion-free case proved in Identity C.37 of the Glossary).
- **Flat connection:** $R = 0$ (∇ is *flat*). Example: ∇^0 on $\mathbb{R}^m$.
- **Transport around loops.** If $(t,s) \mapsto \gamma(t,s)$ is a family of curves and V a field along γ, then $\nabla_{\partial_s\gamma}\nabla_{\partial_t\gamma}V - \nabla_{\partial_t\gamma}\nabla_{\partial_s\gamma}V = R(\partial_s\gamma,\partial_t\gamma,V)$ ([2, Prop. 7.5]): the precise form of the intuition above.

Example 12.1 — the plane in polar coordinates is flat

Use (12.1) with ${\Gamma^{r}}_{\varphi\varphi} = -r$, ${\Gamma^{\varphi}}_{r\varphi} = 1/r$ and compute ${R^{r}}_{r\varphi\varphi}$:

$$ {R^{r}}_{r\varphi\varphi} = \partial_r {\Gamma^{r}}_{\varphi\varphi} - \partial_\varphi {\Gamma^{r}}_{r\varphi} + {\Gamma^{s}}_{\varphi\varphi}{\Gamma^{r}}_{rs} - {\Gamma^{s}}_{r\varphi}{\Gamma^{r}}_{\varphi s} = -1 - 0 + 0 - {\Gamma^{\varphi}}_{r\varphi}{\Gamma^{r}}_{\varphi\varphi} = -1 + \tfrac{1}{r}\cdot r = 0. $$

The Christoffel symbols are nonzero but the curvature vanishes: the nonzero Γ are only an artefact of the coordinates. R **is a tensor; if a tensor vanishes in one chart it vanishes in all charts.**

Example 12.2 — the unit 2-sphere

With ${\Gamma^{\theta}}_{\varphi\varphi} = -\sin\theta\cos\theta$ and ${\Gamma^{\varphi}}_{\theta\varphi} = \cot\theta$:

$$ \begin{aligned} {R^{\theta}}_{\theta\varphi\varphi} &= \partial_\theta {\Gamma^{\theta}}_{\varphi\varphi} - \partial_\varphi {\Gamma^{\theta}}_{\theta\varphi} + {\Gamma^{s}}_{\varphi\varphi}{\Gamma^{\theta}}_{\theta s} - {\Gamma^{s}}_{\theta\varphi}{\Gamma^{\theta}}_{\varphi s} \\ &= -(\cos^2\theta - \sin^2\theta) - 0 + 0 - \cot\theta(-\sin\theta\cos\theta) = \sin^2\theta. \end{aligned} $$

(The only contributing s in the last term is $s = \varphi$.) So $R(\partial_\theta,\partial_\varphi)\partial_\varphi = \sin^2\theta\,\partial_\theta = g_{\varphi\varphi}\,\partial_\theta$, which is the special case of the *constant-curvature formula*

$$ R(X,Y)Z = K\big(g(Y,Z)X - g(X,Z)Y\big), \qquad K = 1 \ \text{(unit sphere)}. $$

For the sphere of radius a, $K = 1/a^2$. The curvature is nonzero in every chart: you cannot flatten a sphere.

12.2 Ricci tensor and scalar curvature

Formal idea: contractions of the curvature

The *Ricci tensor* is the contraction of the contravariant index with the first covariant one, $\mathrm{Ric} = c(R) \in \Pi^0_2(M)$:

$$ R_{jk} = \mathrm{Ric}_{jk} = {R^{i}}_{ijk} = \mathrm{tr}\,\big(v \mapsto R(v,\partial_j)\partial_k\big). $$

When a metric g is given (and $\nabla = \nabla^{LC}$), the *scalar curvature* is the full contraction with the inverse metric

$$s = g^{jk}\,\mathrm{Ric}_{jk} \in C^\infty(M).$$

For ∇^{LC} the Ricci tensor is symmetric.

Intuition: Ricci = "how volumes of small balls differ from flat ones"

R has too many components to be pictured; Ricci averages it over directions. Roughly: a small ball of geodesics converging (positive Ricci) has less volume than in flat space; diverging (negative Ricci), more. In GR the Ricci tensor is sourced by matter: it is what appears in the Einstein equations; the remaining part of R (the *Weyl tensor*) describes gravitational waves and tidal fields in vacuum.

Example 12.3 — Ricci of the unit sphere

$R_{\varphi\varphi} = R^{i}{}_{i\varphi\varphi} = R^{\theta}{}_{\theta\varphi\varphi} + R^{\varphi}{}_{\varphi\varphi\varphi} = \sin^2\theta$ and $R_{\theta\theta} = R^{\varphi}{}_{\varphi\theta\theta}$. Compute the latter: $R^{\varphi}{}_{\varphi\theta\theta} = \partial_\varphi\Gamma^{\varphi}{}_{\theta\theta} - \partial_\theta\Gamma^{\varphi}{}_{\varphi\theta} + \Gamma^{s}{}_{\theta\theta}\Gamma^{\varphi}{}_{\varphi s} - \Gamma^{s}{}_{\varphi\theta}\Gamma^{\varphi}{}_{\theta s} = -\partial_\theta\cot\theta - \cot^2\theta = \csc^2\theta - \cot^2\theta = 1$. Thus

$$\mathrm{Ric} = \mathrm{d}\theta\otimes\mathrm{d}\theta + \sin^2\theta\,\mathrm{d}\varphi\otimes\mathrm{d}\varphi = g, \qquad s = g^{\theta\theta}\,\mathrm{Ric}_{\theta\theta} + g^{\varphi\varphi}\,\mathrm{Ric}_{\varphi\varphi} = 1 + 1 = 2.$$

For the sphere of radius a, with its metric $g = a^2(\mathrm{d}\theta^2 + \sin^2\theta\,\mathrm{d}\varphi^2)$: $\mathrm{Ric} = g/a^2$ (in 2D always $\mathrm{Ric} = \frac{s}{2}\,g$) and $s = 2/a^2 = 2K$. (The components R_{ij} do not depend on a, but g and g^{-1} do.)

12.3 The metric case: the Riemann tensor with all indices down

Formal idea: properties for ∇^{LC}

For a metric g with $\nabla = \nabla^{LC}$ define the totally covariant tensor $\mathbb{R} \in \Pi^0_4(M)$,

$$\mathbb{R}(X,Y,Z,W) = g\big(R(X,Y)Z,W\big), \qquad \mathbb{R}_{ijkl} = g_{lm}R^{m}{}_{ijk}.$$

Besides the general properties,

(a) $\mathbb{R}(X_1,X_2,X_3,X_4) = -\mathbb{R}(X_1,X_2,X_4,X_3)$ (antisymmetry in the last pair);
(b) $\mathbb{R}(X_1,X_2,X_3,X_4) = \mathbb{R}(X_3,X_4,X_1,X_2)$ (pair symmetry);
(c) **(second Bianchi identity)**

$$\nabla_{X_5}\mathbb{R}(X_1,X_2,X_3,X_4) + \nabla_{X_3}\mathbb{R}(X_1,X_2,X_4,X_5) + \nabla_{X_4}\mathbb{R}(X_1,X_2,X_5,X_3) = 0.$$

(a) and (b) are proved in Identity C.38 of the Glossary; for (c) see [2].

Formal idea: how many independent components?

The symmetries and the first Bianchi identity leave $N = \frac{1}{12}m^2(m^2-1)$ independent components:

m	1	2	3	4
N	0	1	6	20

$m = 2$: $N = 1$, so $\mathbb{R}$ is determined by the scalar curvature s (the Gaussian curvature $K = s/2$).

$m = 3$: $N = 6$, and $\mathbb{R}$ is determined by Ric. $m > 3$: $\mathbb{R}$ contains more information than Ric and s (the Weyl tensor); in $m = 4$, 10 components of Weyl + 10 of Ricci = 20.

Formal idea: curvature as a local invariant; normal coordinates

If $\varphi : M \to \tilde{M}$ is a local isometry ($\varphi^* \tilde{g} = g$) then $\varphi^* \tilde{\mathbb{R}} = \mathbb{R}$: curvature is a *local invariant* of the metric. Every point p admits *normal coordinates* (x^i) in which

$$g_{ij}(x) = \delta_{ij} - \tfrac{1}{3}\,\mathbb{R}_{ikjl}(p)\,x^k x^l + O(|x|^3)$$

(and η_{ij} instead of δ_{ij} in the Lorentzian case); in particular $\Gamma(p) = 0$. Conversely $\mathbb{R} \equiv 0$ iff g is locally equivalent to the flat metric.

Intuition: the equivalence principle in mathematical form

At any point you can choose coordinates (a freely falling lab) in which the metric looks like Minkowski up to *second-order* corrections, and these corrections are the curvature. You can erase gravity at a point (first-order effects: $\Gamma = 0$) but not the tidal effects (second-order: R). A small lab sees curvature only through tides.

Exercises

1. Check that (12.1) is antisymmetric in (i, j).
2. Compute $R^{\varphi}{}_{\varphi\theta\theta}$ and $R^{\theta}{}_{\theta\varphi\varphi}$ for a sphere of radius a using $g = a^2(\mathrm{d}\theta^2 + \sin^2\theta\,\mathrm{d}\varphi^2)$ and verify $s = 2/a^2$.
3. For $\mathrm{d}s^2 = -\mathrm{d}t^2 + a(t)^2\mathrm{d}x^2$ compute the only independent component of $\mathbb{R}$ and show that $s = 2\ddot{a}/a$.

13 Geodesic deviation: tidal forces

Prerequisites

11 (Torsion, the Levi-Civita connection and geodesics); 12 (The Riemann curvature tensor)

Intuition: tidal forces

Two nearby freely falling particles (two geodesics) do not stay at a fixed separation if the space is curved: on the sphere two meridians that start parallel at the equator meet at the poles; in the Earth's gravity two stones dropped side by side approach each other as they fall towards the centre; the Moon stretches the oceans. The relative acceleration of nearby geodesics is the physical, measurable face of curvature.

Formal idea: geodesic deviation equation

Let $\gamma_s(t)$ be a family of geodesics (for each s, $t \mapsto \gamma_s(t)$ is a geodesic) and let $\xi = \partial_s \gamma|_{s=0}$ be the *separation vector field* along γ_0. Then, with $D_t = \nabla_{\dot{\gamma}_0}$ the covariant derivative along γ_0

("covariant acceleration" when applied twice),

$$\boxed{D_t D_t\, \xi = R(\dot\gamma_0, \xi)\, \dot\gamma_0}$$

Proof (∇ torsion-free). Write $T = \partial_t \gamma$, $S = \partial_s \gamma$ for the two "velocity" fields of the family $\gamma(t,s)$. Coordinate fields commute, $[T,S] = 0$, and $\nabla_T T = 0$ (each curve is a geodesic). Since the torsion vanishes, $\nabla_T S = \nabla_S T + [T,S] = \nabla_S T$. Then, using the identity $\nabla_T \nabla_S V - \nabla_S \nabla_T V = R(T,S)V$ for fields along a two-parameter family (this is the "curvature = non-commuting derivatives" statement of Section 12, cf. [2, Prop. 7.5]),

$$D_t D_t \xi = \nabla_T \nabla_T S = \nabla_T \nabla_S T = \nabla_S \underbrace{\nabla_T T}_{=0} + R(T,S)T = R(\dot\gamma_0, \xi)\dot\gamma_0 \qquad (s=0).$$

(One can also prove the same statement in a chart by extending $\dot\gamma_0, \xi$ to vector fields on $U \times [t_0, t_1]$ with the connection extended trivially in the t-direction.)

Intuition: curvature = tidal tensor

In coordinates $\frac{D^2 \xi^i}{\mathrm{d}t^2} = R^i{}_{abc}\, \dot\gamma^a \xi^b \dot\gamma^c$: the matrix $\xi \mapsto R(\dot\gamma, \xi)\dot\gamma$ is the **tidal operator**. In the Newtonian limit with potential Φ it becomes $\ddot\xi^i = -\partial_i \partial_j \Phi\, \xi^j$ (the *Newtonian tidal tensor*). Hence "gravity" = geodesic deviation; the force on a single particle is not observable (equivalence principle), the relative force is.

Example 13.1 — two meridians on the unit sphere

For the unit sphere, $R(X,Y)Z = g(Y,Z)X - g(X,Z)Y$. Take a unit-speed geodesic γ_0 and a separation $\xi \perp \dot\gamma_0$ (so $g(\xi, \dot\gamma_0) = 0$, $g(\dot\gamma_0, \dot\gamma_0) = 1$):

$$R(\dot\gamma_0, \xi)\dot\gamma_0 = g(\xi, \dot\gamma_0)\dot\gamma_0 - g(\dot\gamma_0, \dot\gamma_0)\xi = -\xi \quad \Longrightarrow \quad \ddot\xi + \xi = 0.$$

With $\xi(0) = 0$, $\dot\xi(0) = \delta$ (two meridians leaving the north pole at angle δ): $\xi(t) = \delta \sin t$ — the separation grows, reaches its maximum δ at the equator ($t = \pi/2$) and vanishes again at the south pole ($t = \pi$). In flat space ($R = 0$) one would have $\ddot\xi = 0$, $\xi = \delta t$: straight lines diverge linearly. On a negatively curved surface $\ddot\xi = +\xi$ (exponential divergence).

Exercises

1. Check, using $R(X,Y)Z = K(g(Y,Z)X - g(X,Z)Y)$ and (12.1), that the K of a sphere of radius a is $1/a^2$.
2. Show that geodesic deviation forces ξ to stay orthogonal to $\dot\gamma$ if it starts orthogonal with orthogonal derivative.

14 Synthesis III: three derivatives, three invariants

Prerequisites

M1:12 (Tensor fields and differential forms); M1:13 (Integration of 1-forms); M1:16 (The Gauss (divergence) theorem); 7 (The Lie derivative and Cartan's formula); 10 (Connections and the covariant derivative); 11 (Torsion, the Levi-Civita connection and geodesics); 12 (The Riemann curvature tensor)

Synthesis: what this section does

Three different "derivatives" have been built: the exterior derivative d (Section M1:12.3), the Lie derivative $\mathcal{L}_X$ (Section 7) and the covariant derivative ∇ (Section 10). They were introduced separately; here they are compared, the three definitions of divergence are shown to agree, curvature is set against $\mathrm{dd} = 0$, and (in Section CM2:2.2 of Part CM1:I, once Noether's theorem is available) the conservation laws of mechanics are matched with the Killing fields of general relativity.

14.1 The three derivatives side by side

	d	$\mathcal{L}_X$	∇_X
acts on	differential forms only	any tensor field	any tensor field
extra data	none (smooth structure)	the vector field X	a connection (Levi-Civita if a metric is given)
linear in X over functions?	—	no: depends on ∂X	yes: $\nabla_{fX} = f\nabla_X$
second derivative	$\mathrm{dd} = 0$	$[\mathcal{L}_X, \mathcal{L}_Y] = \mathcal{L}_{[X,Y]}$	$[\nabla_X, \nabla_Y] - \nabla_{[X,Y]} = R(X,Y)$
compatible with g?	irrelevant	$\mathcal{L}_X g = 0$ iff X Killing	$\nabla g = 0$ for Levi-Civita

For the Levi-Civita connection they are related by two short statements.

Proposition 14.1. *Let ∇ be the Levi-Civita connection of g (torsion-free, $\nabla g = 0$). Then*

(a) for a 1-form ω: $(\mathrm{d}\omega)(X,Y) = (\nabla_X\omega)(Y) - (\nabla_Y\omega)(X)$, i.e. $(\mathrm{d}\omega)_{ij} = \nabla_i\omega_j - \nabla_j\omega_i = \partial_i\omega_j - \partial_j\omega_i$; the Christoffel symbols cancel;

(b) for the metric: $(\mathcal{L}_X g)_{ij} = \nabla_i X_j + \nabla_j X_i$; in particular X is Killing iff $\nabla_i X_j$ is antisymmetric.

Proof. (a) $(\nabla_X\omega)(Y) = X(\omega(Y)) - \omega(\nabla_X Y)$. Subtract the same expression with X, Y exchanged and use $\nabla_X Y - \nabla_Y X = [X,Y]$ (torsion-free): the result is $X(\omega(Y)) - Y(\omega(X)) - \omega([X,Y]) = \mathrm{d}\omega(X,Y)$ by the invariant formula of Theorem M1:12.5. (b) $\mathcal{L}_X g(Y,Z) = X(g(Y,Z)) - g([X,Y],Z) - g(Y,[X,Z])$. By $\nabla g = 0$, $X(g(Y,Z)) = g(\nabla_X Y, Z) + g(Y, \nabla_X Z)$, and $[X,Y] = \nabla_X Y - \nabla_Y X$; substituting, $\mathcal{L}_X g(Y,Z) = g(\nabla_Y X, Z) + g(Y, \nabla_Z X)$. □

So d is precisely the part of ∇ that does not see the connection: the antisymmetrisation of $\nabla\omega$ is independent of Γ.

14.2 Divergence three ways

Divergence has appeared in four guises:

Definition	Formula	Needs
rate of change of volume	$\mathcal{L}_X\mu = (\operatorname{div}_\mu X)\mu$, $\mathrm{d}(\iota_X\mu) = \operatorname{div}_\mu X\,\mu$	a volume form μ
coordinates	$\frac{1}{\sqrt{\lvert g\rvert}}\partial_i(\sqrt{\lvert g\rvert}X^i)$	chart, g
trace of the covariant derivative	$\nabla_i X^i = \partial_i X^i + \Gamma^i_{ij}X^j$	Levi-Civita connection
geometric algebra	$\boldsymbol{\nabla} \rfloor X = \boldsymbol{\nabla}\cdot X$	Q

Proposition 14.2. *For the Levi-Civita connection,* $\Gamma^i_{ij} = \partial_j \ln\sqrt{\lvert g\rvert}$, *hence* $\nabla_i X^i = \dfrac{1}{\sqrt{\lvert g\rvert}}\,\partial_i\big(\sqrt{\lvert g\rvert}\,X^i\big) = \operatorname{div}_{\mathrm{vol}_g} X$.

Proof. From the formula (11.1), $\Gamma^i_{ij} = \frac{1}{2}g^{il}(\partial_i g_{jl} + \partial_j g_{il} - \partial_l g_{ij})$. The first and third terms cancel (rename $i \leftrightarrow l$ in the third and use the symmetry of g), leaving $\frac{1}{2}g^{il}\partial_j g_{il} = \partial_j \ln\sqrt{\lvert g\rvert}$ by Jacobi's formula $\partial_j \ln\lvert\det g\rvert = g^{il}\partial_j g_{il}$. Then $\nabla_i X^i = \partial_i X^i + X^j\partial_j \ln\sqrt{\lvert g\rvert} = \frac{1}{\sqrt{\lvert g\rvert}}\partial_i(\sqrt{\lvert g\rvert}X^i)$. □

Example 14.3 — Polar coordinates: the extra term is Γ

With $g = \operatorname{diag}(1, r^2)$, $\sqrt{\lvert g\rvert} = r$ and $\Gamma^i_{ir} = \Gamma^\varphi_{\varphi r} = 1/r$, Example 10.3 gave $\nabla_i V^i = \frac{1}{r}\partial_r(rV^r) + \partial_\varphi V^\varphi$: the naive $\partial_i V^i$ misses the term V^r/r, and the volume-form definition $\frac{1}{\sqrt{g}}\partial_i(\sqrt{g}V^i)$ produces it automatically. For the position field $V = r\partial_r$: $\operatorname{div} V = \frac{1}{r}\partial_r(r^2) = 2$, the dimension of the plane, as in Example M1:16.3 (where the answer in $\mathbb{R}^3$ is 3).

14.3 Curvature versus dd = 0

Synthesis: the same "loop per unit area" principle, one level up

For a 1-form ω, Section M1:13.6 showed that $\mathrm{d}\omega(u,v)\,\varepsilon\delta$ is the *circulation* $\oint\omega$ around the small parallelogram spanned by $\varepsilon u, \delta v$: it measures how far the sheets of ω fail to close up. For a connection, the curvature $R(u,v)w\,\varepsilon\delta$ is the *rotation* of w after parallel transport around the same parallelogram (Section 12): it measures how far parallel transport fails to be path independent. In both cases a loop integral per unit area is a second derivative, with the *antisymmetrisation* of mixed derivatives as its algebraic form. The difference is exactly this: partial derivatives commute, $\partial_i\partial_j - \partial_j\partial_i = 0$, hence $\mathrm{dd} = 0$ (Example M1:12.7); covariant derivatives do not,

$$[\nabla_i, \nabla_j]V^k = R^k{}_{lij}V^l.$$

The local counterparts of the two vanishings are also parallel: $\mathrm{d}\omega = 0 \Rightarrow \omega = \mathrm{d}f$ locally (Poincaré lemma), and $R = 0 \Rightarrow$ locally there are coordinates with g_{ij} constant. On a sphere $R \neq 0$: transporting a vector around the octant triangle of Example 11.2 turns it by $\pi/2$, the analogue of the angle form being closed but not exact (Example M1:13.4) – except that here the obstruction is local, not topological.

Key points of Part IV

- A connection makes vectors at nearby points comparable; ∇_X is C^∞-linear in X, unlike $\mathcal{L}_X$. Levi-Civita: torsion-free and $\nabla g = 0$; it is determined by g alone.
- d is the connection-independent antisymmetrisation of ∇; the three definitions of diver-

gence coincide; $\mathcal{L}_X g = 0 \iff \nabla_{(i} X_{j)} = 0$.

- Curvature is the failure of ∇-derivatives to commute: the analogue of $\mathrm{dd} = 0$ with the answer *not* zero.

15 Lie groups and Lie algebras

TODO

TODO(M2/ch15): this chapter has no content yet. Its place in the reading order is listed in `CHAPTER_ORGANIZATION.md`.

Prerequisites: TODO(M2/ch06); 7 (The Lie derivative and Cartan's formula)

16 Fibre bundles, gauge connections and spinors

TODO

TODO(M2/ch16): this chapter has no content yet. Its place in the reading order is listed in `CHAPTER_ORGANIZATION.md`.

Prerequisites: 10 (Connections and the covariant derivative); TODO(M2/ch15)

Part IV Complex analysis: the residue theorem as Stokes' theorem

17 The residue theorem as Stokes' theorem

Prerequisites

M1:9 (The exterior algebra and the wedge product); M1:13 (Integration of 1-forms); M1:14 (From 1-forms to k-forms: integration is full contraction); M1:15 (Stokes' theorem as it is understood today); 1 (The geometric (Clifford) algebra); 4 (Duality: the pseudoscalar, the dual, the cross product and the Hodge star); 8 (The exterior derivative in geometric algebra); 9 (Synthesis II: one calculus, three languages)

Is the residue theorem related to all of this? Yes: it is not a separate theorem but Stokes' theorem applied to a particular kind of complex-valued 1-form on a region with holes. Cauchy's theorem is Stokes for a closed form; the residue theorem is Stokes for a closed form that fails to be exact exactly at the punctures, which is the mechanism of Example M1:13.4 (the angle form $\mathrm{d}\theta$ on $\mathbb{R}^2 \setminus \{0\}$). Its real and imaginary parts are, respectively, the circulation and the flux of Sections M1:15–CM1:3. For the classical theory see [16]; for a geometric account, [17].

17.1 Complex-valued 1-forms and the contraction

Identify $\mathbb{C} = \mathbb{R}^2$ with $z = x + iy$. The form $\mathrm{d}z = \mathrm{d}x + i\,\mathrm{d}y$ is a complex-valued 1-form (a $\mathbb{C}$-valued covector at each point), and its contraction with a tangent vector $v = (v^x, v^y) \simeq v^x + iv^y$ is simply $\mathrm{d}z(v) = v$. For a function $f : U \subset \mathbb{C} \to \mathbb{C}$ the form $f\,\mathrm{d}z$ contracts with a velocity as $\langle f\,\mathrm{d}z, \dot\gamma(t)\rangle = f(\gamma(t))\dot\gamma(t)$,

$$\int_\gamma f\,\mathrm{d}z = \int_a^b f(\gamma(t))\,\dot\gamma(t)\,\mathrm{d}t, \tag{17.1}$$

which is Definition (M1:13.1) extended $\mathbb{C}$-linearly. For every smooth f one has $\mathrm{d}f = \partial_z f\,\mathrm{d}z + \partial_{\bar z} f\,\mathrm{d}\bar z$, and

$$f \text{ holomorphic} \iff \partial_{\bar z} f = 0 \iff \mathrm{d}f = f'(z)\,\mathrm{d}z \iff \mathrm{d}f_p : \mathbb{R}^2 \to \mathbb{C} \text{ is } \mathbb{C}\text{-linear.} \tag{17.2}$$

The Cauchy–Riemann equations are thus the statement that the covector $\mathrm{d}f$ is complex-linear: $\mathrm{d}f(iv) = i\,\mathrm{d}f(v)$.

Steps behind these statements. With $\partial_z = \tfrac12(\partial_x - i\partial_y)$ and $\partial_{\bar z} = \tfrac12(\partial_x + i\partial_y)$:

$$\begin{aligned}\partial_z f\,\mathrm{d}z + \partial_{\bar z} f\,\mathrm{d}\bar z &= \tfrac12(f_x - if_y)(\mathrm{d}x + i\,\mathrm{d}y) + \tfrac12(f_x + if_y)(\mathrm{d}x - i\,\mathrm{d}y)\\ &= \tfrac12\big(f_x\mathrm{d}x + if_x\mathrm{d}y - if_y\mathrm{d}x + f_y\mathrm{d}y\big) + \tfrac12\big(f_x\mathrm{d}x - if_x\mathrm{d}y + if_y\mathrm{d}x + f_y\mathrm{d}y\big)\\ &= f_x\mathrm{d}x + f_y\mathrm{d}y = \mathrm{d}f.\end{aligned}$$

Writing $f = u + iv$, the condition $\partial_{\bar z} f = 0$ means $f_x + if_y = (u_x - v_y) + i(v_x + u_y) = 0$, *i.e.* the usual Cauchy–Riemann equations $u_x = v_y$, $u_y = -v_x$. Finally $\mathrm{d}f = f'\mathrm{d}z$ means $\partial_{\bar z} f = 0$ and $\partial_z f = f'$.

17.2 Cauchy's theorem and the Cauchy–Riemann equations as "$\mathrm{div} = 0$, $\mathrm{curl} = 0$"

Let us compute $\mathrm{d}(f\,\mathrm{d}z)$ in three steps.

1. Leibniz rule and $\mathrm{d}(\mathrm{d}z) = 0$: $\mathrm{d}(f\,\mathrm{d}z) = \mathrm{d}f \wedge \mathrm{d}z$.
2. Insert $\mathrm{d}f = \partial_z f\,\mathrm{d}z + \partial_{\bar z} f\,\mathrm{d}\bar z$ and use $\mathrm{d}z \wedge \mathrm{d}z = 0$: $\mathrm{d}f \wedge \mathrm{d}z = \partial_{\bar z} f\,\mathrm{d}\bar z \wedge \mathrm{d}z$.
3. Expand $\mathrm{d}\bar z \wedge \mathrm{d}z = (\mathrm{d}x - i\,\mathrm{d}y) \wedge (\mathrm{d}x + i\,\mathrm{d}y) = i\,\mathrm{d}x \wedge \mathrm{d}y - i\,\mathrm{d}y \wedge \mathrm{d}x = 2i\,\mathrm{d}x \wedge \mathrm{d}y$.

$$\mathrm{d}(f\,\mathrm{d}z) = \partial_{\bar z} f\,\mathrm{d}\bar z \wedge \mathrm{d}z = 2i\,\partial_{\bar z} f\,\mathrm{d}x \wedge \mathrm{d}y. \tag{17.3}$$

Hence f is holomorphic if and only if the 1-form $f\,\mathrm{d}z$ is closed, and Stokes' theorem gives at once

Theorem 17.1 (Cauchy). *If f is holomorphic on a neighbourhood of a compact region $\Omega \subset \mathbb{C}$ with piecewise smooth boundary, then $\oint_{\partial\Omega} f\,\mathrm{d}z = \int_\Omega \mathrm{d}(f\,\mathrm{d}z) = 0$.*

Real and imaginary parts: circulation and flux

Write $w = u - iv$ (the "complex velocity" of a planar vector field $\mathbf{u} = (u, v)$). Then

$$w\,\mathrm{d}z = \underbrace{(u\,\mathrm{d}x + v\,\mathrm{d}y)}_{\mathbf{u}^\flat} + i\,\underbrace{(u\,\mathrm{d}y - v\,\mathrm{d}x)}_{\iota_{\mathbf{u}}(\mathrm{d}x\wedge\mathrm{d}y) = \star\mathbf{u}^\flat}. \tag{17.4}$$

The real part is the covector $\mathbf{u}^\flat$ (its integral is the circulation Γ), and the imaginary part is the contraction $\iota_{\mathbf{u}}\mathrm{vol}$ (its integral is the flux Q through the curve, cf. (M1:14.2)). The split follows from $w\,\mathrm{d}z = (u - iv)(\mathrm{d}x + i\,\mathrm{d}y) = u\,\mathrm{d}x + iu\,\mathrm{d}y - iv\,\mathrm{d}x + v\,\mathrm{d}y$ and from $\iota_{\mathbf{u}}(\mathrm{d}x \wedge \mathrm{d}y) = u\,\mathrm{d}y - v\,\mathrm{d}x$ (Example M1:9.19). Moreover, from (17.3), in steps:

$$\partial_{\bar z} w = \tfrac{1}{2}(\partial_x + i\partial_y)(u - iv) = \tfrac{1}{2}\big[(u_x + v_y) + i(u_y - v_x)\big] = \tfrac{1}{2}\big(\operatorname{div}\mathbf{u} - i\operatorname{curl}\mathbf{u}\big)$$

(with $\operatorname{curl}\mathbf{u} = \partial_x v - \partial_y u$), so *$w$ is holomorphic $\iff$ the flow is incompressible* ($\operatorname{div}\mathbf{u} = 0$) *and irrotational* ($\operatorname{curl}\mathbf{u} = 0$). Cauchy's theorem is then Green's theorem (Kelvin–Stokes in the plane) for the real part and the divergence theorem (Gauss) for the imaginary part, both applied at the same time.

Special case 17.2 — the Hodge star in dimension two

In the language of Section 9.4, in dimension two the Hodge star on 1-forms satisfies $\star\mathrm{d}z = -i\,\mathrm{d}z$ and $\star\mathrm{d}\bar z = i\,\mathrm{d}\bar z$: $\mathrm{d}z$ is an eigenform of $\star$, and since $\star$ on 1-forms depends only on the conformal class of the metric, holomorphy is a conformally invariant notion (conformal maps preserve holomorphic functions). The eigenform property is checked in one line: $\star\mathrm{d}x = \mathrm{d}y$, $\star\mathrm{d}y = -\mathrm{d}x$, so $\star\mathrm{d}z = \mathrm{d}y - i\,\mathrm{d}x = -i(\mathrm{d}x + i\,\mathrm{d}y) = -i\,\mathrm{d}z$.

17.3 The residue theorem

Let f be holomorphic on an open set containing $\overline{\Omega}$ except at finitely many points $a_1, \dots, a_k \in \Omega$ where it has isolated singularities (poles or essential singularities). The *residue* of f at a is the coefficient c_{-1} of its Laurent expansion $f(z) = \sum_n c_n (z - a)^n$, equivalently

$$\operatorname{Res}_a f = \frac{1}{2\pi i} \oint_{|z-a|=\varepsilon} f\,\mathrm{d}z \qquad (\text{small } \varepsilon > 0,\ \text{counter-clockwise}). \tag{17.5}$$

Why the two definitions agree (steps). Parametrise the circle: $z = a + \varepsilon e^{it}$, $t \in [0, 2\pi]$, so $\mathrm{d}z = i\varepsilon e^{it}\,\mathrm{d}t$. For an integer n:

$$\oint (z-a)^n\,\mathrm{d}z = \int_0^{2\pi} \varepsilon^n e^{int}\, i\varepsilon e^{it}\,\mathrm{d}t = i\varepsilon^{n+1}\int_0^{2\pi} e^{i(n+1)t}\,\mathrm{d}t = \begin{cases} 2\pi i, & n = -1, \\ 0, & n \neq -1, \end{cases}$$

because for $m = n+1 \neq 0$, $\int_0^{2\pi} e^{imt}\,\mathrm{d}t = \left[e^{imt}/(im)\right]_0^{2\pi} = 0$, while for $m = 0$ the integrand is 1. Integrating the Laurent series term by term (it converges uniformly on the circle) leaves only $c_{-1} \cdot 2\pi i$.

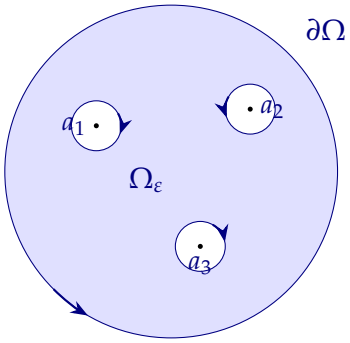


Figure 2: Proof of the residue theorem. Stokes on $\Omega_\varepsilon = \Omega$ minus small discs: the outer boundary runs counter-clockwise and the small circles clockwise (region on the left). Since $f\,\mathrm{d}z$ is closed on Ω_ε, the integral over the outer boundary equals the sum of the integrals over the small circles traversed counter-clockwise. This is the annulus of Figure M1:3 with several holes.

Theorem 17.3 (Residue theorem)**.** *Under these assumptions,*

$$\oint_{\partial\Omega} f\,\mathrm{d}z = 2\pi i \sum_{j=1}^{k} \operatorname{Res}_{a_j} f.$$

Proof. Remove from Ω small open discs $D_j = D(a_j, \varepsilon)$ and let $\Omega_\varepsilon = \Omega \setminus \bigcup_j D_j$, a manifold with boundary $\partial\Omega_\varepsilon = \partial\Omega - \sum_j \partial D_j$ (the circles are traversed clockwise: induced orientation, Figure 2). On Ω_ε the form $f\,\mathrm{d}z$ is smooth and closed, so Stokes' theorem (M1:15.1) gives

$$0 = \int_{\Omega_\varepsilon} \mathrm{d}(f\,\mathrm{d}z) = \oint_{\partial\Omega} f\,\mathrm{d}z - \sum_j \oint_{\partial D_j^{\mathrm{ccw}}} f\,\mathrm{d}z,$$

and each small-circle integral is $2\pi i \operatorname{Res}_{a_j} f$ by (17.5). □

Example 17.4 — Two applications of the residue theorem

(a) $\displaystyle\oint_{|z|=2} \frac{\mathrm{d}z}{z(z-1)}$. Partial fractions: $\frac{1}{z(z-1)} = \frac{1}{z-1} - \frac{1}{z}$, so $\operatorname{Res}_1 = +1$ and $\operatorname{Res}_0 = -1$ (both poles lie inside $|z| < 2$). The theorem gives $2\pi i(1-1) = 0$. *Check:* $\oint \frac{\mathrm{d}z}{z-1} = 2\pi i$ and $\oint \frac{\mathrm{d}z}{z} = 2\pi i$ by the circle computation above, and their difference is 0.

(b) $\displaystyle\oint_{|z|=1} \frac{e^z}{z^2}\,\mathrm{d}z$. Expand $e^z = 1 + z + \frac{z^2}{2} + \cdots$, so $\frac{e^z}{z^2} = \frac{1}{z^2} + \frac{1}{z} + \frac{1}{2} + \cdots$ and $c_{-1} = 1$. The integral is $2\pi i \cdot 1 = 2\pi i$. (Equivalently, $\operatorname{Res}_0 = \frac{\mathrm{d}}{\mathrm{d}z} e^z\big|_0 = 1$, the formula for a double pole.)

why only c_{-1} survives: exact versus closed

On $\mathbb{C} \setminus \{a\}$ the forms $(z-a)^n \mathrm{d}z$ with $n \neq -1$ are *exact*: $(z-a)^n \mathrm{d}z = \mathrm{d}\left(\frac{(z-a)^{n+1}}{n+1}\right)$, so they integrate to zero on any closed loop (Section M1:13.5). The only closed, non-exact one is

$$\frac{\mathrm{d}z}{z-a} = \mathrm{d}\ln|z-a| + i\,\mathrm{d}\theta, \tag{17.6}$$

whose imaginary part is precisely the angle form $\mathrm{d}\theta$ of Example M1:13.4, with $\oint \mathrm{d}\theta = 2\pi$; hence $\oint \frac{\mathrm{d}z}{z-a} = i \cdot 2\pi = 2\pi i$. Derivation of the identity: write $z = re^{i\theta}$; then $\mathrm{d}z = e^{i\theta}\mathrm{d}r + ire^{i\theta}\mathrm{d}\theta$ and $\frac{\mathrm{d}z}{z} = \frac{\mathrm{d}r}{r} + i\,\mathrm{d}\theta = \mathrm{d}\ln r + i\,\mathrm{d}\theta$. In other words, the residue measures the component of $f\,\mathrm{d}z$ along the one cohomology class which is not exact near a.

17.4 The cohomological meaning of the residue

For $X = \mathbb{C} \setminus \{a_1, \dots, a_k\}$ one has $H^1_{\mathrm{dR}}(X;\mathbb{C}) \cong \mathbb{C}^k$, generated by the classes of $\mathrm{d}z/(z-a_j)$, and $H_1(X;\mathbb{Z}) \cong \mathbb{Z}^k$, generated by small circles around the a_j. The residue is the de Rham pairing between these [3]: $\operatorname{Res}_{a_j} f = \frac{1}{2\pi i}\langle [\text{circle around } a_j], [f\,\mathrm{d}z]\rangle$. A closed loop γ in X is homologous to $\sum_j n_j[\text{circle}_j]$, where $n_j = \frac{1}{2\pi i}\oint_\gamma \frac{\mathrm{d}z}{z-a_j}$ is the winding number of γ around a_j (Example M1:13.4); *e.g.* a loop that goes twice around a_j has $\oint \frac{\mathrm{d}z}{z-a_j} = 2 \cdot 2\pi i$, so $n_j = 2$. This yields the general version

$$\oint_\gamma f\,\mathrm{d}z = 2\pi i \sum_j n(\gamma, a_j)\operatorname{Res}_{a_j} f. \tag{17.7}$$

The integral depends only on the homology class of the contour: a contour can be deformed freely as long as it does not cross a singularity, which is the practical principle behind contour integration.

17.5 Beyond holomorphic functions: Cauchy–Pompeiu and the Coulomb law in 2D

If f is merely smooth, Stokes still applies. Let us derive the Cauchy–Pompeiu formula step by step, with $\omega = \dfrac{f\,\mathrm{d}z}{z-z_0}$ on $\Omega_\varepsilon = \Omega \setminus D(z_0,\varepsilon)$.

1. *Differentiate.* $1/(z-z_0)$ is holomorphic away from z_0, so $\partial_{\bar z}\big(f/(z-z_0)\big) = \partial_{\bar z} f/(z-z_0)$ and, by (17.3), $\mathrm{d}\omega = \dfrac{\partial_{\bar z} f}{z-z_0}\, 2i\,\mathrm{d}x\wedge\mathrm{d}y$.
2. *Stokes on* Ω_ε (outer boundary ccw, small circle cw): $\displaystyle\oint_{\partial\Omega}\omega - \oint_{|z-z_0|=\varepsilon}\omega = \int_{\Omega_\varepsilon}\mathrm{d}\omega$.
3. *The small circle.* With $z = z_0 + \varepsilon e^{it}$: $\displaystyle\oint\omega = \int_0^{2\pi}\frac{f(z_0+\varepsilon e^{it})}{\varepsilon e^{it}}\, i\varepsilon e^{it}\,\mathrm{d}t = i\int_0^{2\pi} f(z_0+\varepsilon e^{it})\,\mathrm{d}t \xrightarrow[\varepsilon\to 0]{} 2\pi i f(z_0)$ by continuity of f.
4. *Let* $\varepsilon \to 0$. The area integral converges because $1/|z-z_0|$ is integrable in the plane (area element $r\,\mathrm{d}r\,\mathrm{d}\theta$). We get $\displaystyle\oint_{\partial\Omega}\frac{f\,\mathrm{d}z}{z-z_0} - 2\pi i f(z_0) = 2i\iint_\Omega \frac{\partial_{\bar z} f}{z-z_0}\,\mathrm{d}A$.
5. *Divide by* $2\pi i$ *and solve for* $f(z_0)$*:*

$$f(z_0) = \frac{1}{2\pi i}\oint_{\partial\Omega}\frac{f(z)}{z-z_0}\,\mathrm{d}z - \frac{1}{\pi}\iint_\Omega \frac{\partial_{\bar z} f}{z-z_0}\,\mathrm{d}A. \tag{17.8}$$

For holomorphic f the area term vanishes and we recover Cauchy's integral formula (for (17.8) and $\bar\partial$ see [19]). In the language of distributions the same computation says $\partial_{\bar z}\big(\frac{1}{\pi z}\big) = \delta_0$, *i.e.* $1/(\pi z)$ is the fundamental solution of $\partial_{\bar z}$, in perfect analogy with the Gauss law in the plane (Section GR:3.1): a point charge Q (or a point vortex with circulation Γ and source Q) is a simple pole of the complex field w whose residue encodes the charge: $\operatorname{Res}_0 w = \frac{\Gamma+iQ}{2\pi i}$, *i.e.* $2\pi i \operatorname{Res}_0 w = \Gamma + iQ$,

$$w(z) = \frac{\Gamma+iQ}{2\pi i}\,\frac{1}{z} \quad\Longrightarrow\quad \oint w\,\mathrm{d}z = \Gamma + iQ.$$

This is why the residue theorem reads, in electrostatics or fluid mechanics, as "the total flux (and circulation) through a curve equals the sum of the charges (vortices) inside".

17.6 The argument principle: counting zeros with a 1-form

Let f be meromorphic on a neighbourhood of the closure of Ω, with no zeros or poles on $\partial\Omega$. Then $\frac{f'}{f}\mathrm{d}z = \mathrm{d}\log f = \mathrm{d}\ln|f| + i\,\mathrm{d}\arg f = f^*\left(\frac{\mathrm{d}z}{z}\right)$. The first term is exact, so the residue theorem becomes

$$\frac{1}{2\pi i}\oint_{\partial\Omega}\frac{f'}{f}\,\mathrm{d}z = \frac{1}{2\pi}\oint_{\partial\Omega} f^*\mathrm{d}\theta = N - P, \tag{17.9}$$

where N and P are the numbers of zeros and poles in Ω counted with multiplicity: the winding number of the curve $f(\partial\Omega)$ around 0. *Steps.* (1) Locally $\log f = \ln|f| + i\arg f$, and $\mathrm{d}\log f = \frac{\mathrm{d}f}{f} = \frac{f'}{f}\mathrm{d}z$; this is also the pull-back $f^*(\mathrm{d}z/z)$ because $f^*\mathrm{d}z = \mathrm{d}f = f'\mathrm{d}z$. (2) $\mathrm{d}\ln|f|$ is exact, so its integral over a closed curve is 0. (3) What remains is $i\oint \mathrm{d}\arg f = 2\pi i\cdot(\text{winding number of } f\circ\gamma \text{ around } 0)$. (4) By the residue theorem, $\frac{f'}{f}$ has a simple pole with residue $+m$ at a zero of order m and $-m$ at a pole of order m (since $f = (z-a)^{\pm m}g$ gives $f'/f = \pm m/(z-a) + g'/g$). *Numerical check.* $f(z) = z^2$ on $|z| = 1$: $\frac{f'}{f} = \frac{2}{z}$, $\frac{1}{2\pi i}\oint\frac{2}{z}\,\mathrm{d}z = 2 = N$ (a double zero at the origin), and $f(e^{it}) = e^{2it}$ winds twice around 0. This is the degree of the map $f|_{\partial\Omega}:\partial\Omega\to\mathbb{C}\setminus\{0\}$, computed by pulling back the angle form (Section M1:15), and it is the basis of Rouché's theorem and of the fundamental theorem of algebra.

Example 17.5 — A real integral by residues, step by step

To compute $I = \displaystyle\int_{-\infty}^{\infty}\frac{\mathrm{d}x}{1+x^2}$, apply Theorem 17.3 to $f\,\mathrm{d}z$ with $f = \dfrac{1}{1+z^2}$ on the upper half-disc Ω_R of radius $R > 1$, whose boundary is the segment $[-R,R]$ followed by the semicircle C_R.

1. *Poles.* $1+z^2 = (z-i)(z+i)$; only $z = i$ lies inside Ω_R.
2. *Residue.* $\operatorname{Res}_i f = \lim_{z\to i}\frac{z-i}{(z-i)(z+i)} = \frac{1}{2i}$.
3. *Theorem.* $\int_{-R}^{R} f(x)\,\mathrm{d}x + \int_{C_R} f\,\mathrm{d}z = 2\pi i\cdot\frac{1}{2i} = \pi$. (On the real axis the pull-back of $f\,\mathrm{d}z$ is $f(x)\,\mathrm{d}x$, the contraction with the velocity 1.)
4. *The arc vanishes.* On $|z| = R$, $|1+z^2| \geq R^2 - 1$, so $\left|\int_{C_R} f\,\mathrm{d}z\right| \leq \dfrac{\pi R}{R^2-1} \to 0$ as $R\to\infty$ (length πR times the bound on $|f|$).
5. *Limit.* Letting $R\to\infty$: $I = \pi$.

Check: $\int\frac{\mathrm{d}x}{1+x^2} = \arctan x\big|_{-\infty}^{\infty} = \frac{\pi}{2} - \left(-\frac{\pi}{2}\right) = \pi$ ✓.

17.7 The current picture

- *Riemann surfaces.* If η is a meromorphic 1-form on a compact Riemann surface S (no boundary), Stokes with small discs removed gives $\sum_{p\in S}\operatorname{Res}_p\eta = 0$: the sum of all residues vanishes. Combined with Poincaré/Serre duality this is a key ingredient in the proofs of the Riemann–Roch theorem and in the theory of abelian differentials. Note that this statement does not even need a metric: η is a 1-form and $\operatorname{Res}_p\eta$ is intrinsic, which is why residues are defined for forms $f\,\mathrm{d}z$ and not for functions f (the residue of the function f changes under a change of coordinate $z\mapsto h(z)$, that of the form $f\,\mathrm{d}z$ does not).

- *Several complex variables.* The kernel $1/(z-z_0)$ is replaced by the Bochner–Martinelli kernel, and residues by Poincaré–Leray residues of closed forms with poles along hypersurfaces; both are proved with Stokes' theorem and are expressed through de Rham cohomology of the complement of the singular set.
- *Currents.* Formulae such as $\partial_{\bar z}(1/\pi z)=\delta_0$ and the Poincaré–Lelong formula $\frac{i}{\pi}\partial\bar\partial\log|f| = [\{f=0\}]$ treat the residue as the current of integration over the singular set (Section 20, item 3), exactly as Stokes' theorem is treated in geometric measure theory [18].

Complex analysis	**Differential forms**
f holomorphic	$f\,\mathrm{d}z$ closed; $\mathrm{d}f$ is $\mathbb{C}$-linear
Cauchy's theorem	Stokes for a closed form on Ω
Cauchy–Riemann equations	$\operatorname{div}=0$ and $\operatorname{curl}=0$ (real/imaginary parts of $w\,\mathrm{d}z$)
Residue $\operatorname{Res}_a f$	period of $f\,\mathrm{d}z$ over a small circle = de Rham pairing
Residue theorem	Stokes on Ω minus discs; boundary = outer curve − small circles
Winding number, argument principle	$\frac{1}{2\pi}\oint f^*\mathrm{d}\theta$ = degree of a map to S^1
Cauchy–Pompeiu, $\partial_{\bar z}(1/\pi z)=\delta$	Stokes with $\mathrm{d}\omega\neq 0$; Gauss law in the plane

17.8 Synthesis IV: complex analysis in the geometric algebra of the plane

Synthesis: where does the imaginary unit come from?

The symbol i has appeared in three guises. (1) As a scalar: $\mathbb{C}=\mathrm{Cl}(0,1)$, the geometric algebra of a line with $e^2=-1$ (Section 1). (2) As a bivector: in the plane algebra $\mathrm{Cl}(2,0)$ the unit area element $I=e_1e_2$ satisfies $I^2=-1$, and the even subalgebra $\{x+Iy\}$ is $\mathbb{C}$; multiplying a vector on the left by e_1 identifies $\mathbb{R}^2$ with $\mathbb{C}$, $e_1(x+Iy)=xe_1+ye_2$. (3) As the factor in complex-valued forms such as $\mathrm{d}z=\mathrm{d}x+i\,\mathrm{d}y$, where i is a formal scalar commuting with everything. In option (2) i is the *oriented plane itself*, and "multiplication by i" is a quarter-turn *because* I rotates vectors by 90° under the geometric product: the dual of a vector in 2D is the vector turned by a quarter turn (Section 4).

Proposition 17.6 (Cauchy–Riemann as a single multivector equation)**.** *Let* $\mathbf{u}=u\,e_1+v\,e_2$ *be a vector field in the plane,* $\boldsymbol{\nabla}=e_1\partial_x+e_2\partial_y$, $I=e_1e_2$, *and* $w=u-iv$ *the complex velocity of* (17.4). *Then*

$$\boldsymbol{\nabla}\,\mathbf{u}=\operatorname{div}\mathbf{u}+I\,\operatorname{curl}\mathbf{u}=2\,\partial_z\bar w,$$

so w *is holomorphic if and only if* $\boldsymbol{\nabla}\mathbf{u}=0$ *(a* monogenic *function).*

Proof. $\boldsymbol{\nabla}\mathbf{u}=(e_1\partial_x+e_2\partial_y)(ue_1+ve_2)=(u_x+v_y)+e_1e_2(v_x-u_y)$, using $e_1e_2=-e_2e_1$ and $e_i^2=1$; this is $\operatorname{div}\mathbf{u}+I\operatorname{curl}\mathbf{u}$. On the other hand $2\partial_z\bar w=(\partial_x-i\partial_y)(u+iv)=(u_x+v_y)+i(v_x-u_y)$ with $i\leftrightarrow I$. Holomorphy of w means $\partial_{\bar z}w=0$, whose conjugate is $\partial_z\bar w=0$. □

Example 17.7 — a monogenic field

$w=z^2=(x^2-y^2)+2ixy$. Since $w=u-iv$, we read off $u=x^2-y^2$ and $v=-2xy$. Then $\operatorname{div}\mathbf{u}=2x-2x=0$ and $\operatorname{curl}\mathbf{u}=v_x-u_y=-2y+2y=0$, so $\boldsymbol{\nabla}\mathbf{u}=0$ ✓.

Synthesis: Cauchy's theorem, three ways

Forms: $\mathrm{d}(f\,\mathrm{d}z) = 2i\,\partial_{\bar z} f\,\mathrm{d}x\wedge\mathrm{d}y$ (equation (17.3)) and Stokes. *Vector calculus:* Green's theorem for the real part and the divergence theorem for the imaginary part of (17.4). *Geometric algebra:* the single equation $\nabla\mathbf{u} = 0$ of Proposition 17.6; in higher dimension the same equation $\nabla F = 0$ defines monogenic functions (Clifford analysis), for which an analogue of Cauchy's integral formula holds (Hestenes–Sobczyk [22]).

18 Hilbert spaces, distributions and spectral theory

TODO

TODO(M2/ch18): this chapter has no content yet. Its place in the reading order is listed in `CHAPTER_ORGANIZATION.md`.

Prerequisites: TODO(M1/ch17); 17 (The residue theorem as Stokes' theorem)

19 Stochastic processes and statistical methods

TODO

TODO(M2/ch19): this chapter has no content yet. Its place in the reading order is listed in `CHAPTER_ORGANIZATION.md`.

Prerequisites: TODO(M1/ch20); TODO(M2/ch18)

Part V Perspective: current state and master summary

20 The current state of the subject

Prerequisites

M1:15 (Stokes' theorem as it is understood today); 1 (The geometric (Clifford) algebra); 9 (Synthesis II: one calculus, three languages); 17 (The residue theorem as Stokes' theorem)

The statements above are the standard, textbook ("smooth") form of the theorems (Spivak, Lee, Bott–Tu [1, 2, 3]). Active and established developments extend them in several directions.

1. **Conceptual unification.** Since the early twentieth century (Poincaré, Cartan, de Rham) the gradient theorem, Green, Kelvin–Stokes and Gauss are viewed as one theorem, (M1:15.1), equivalent to the duality (M1:15.2). The 19th-century formulations (Gauss, Green, Ostrogradsky, Kelvin, Stokes) are recovered by the dictionary $\flat, \sharp, \star$ (Section 9.4); the geometric-algebra formulation of the same dictionary is Section 9.

2. **Rough domains and non-smooth data.** Stokes holds for manifolds with corners and piecewise smooth chains, and for Lipschitz domains (with the normal defined almost everywhere and traces of Sobolev fields). The De Giorgi–Federer divergence theorem holds for sets E of finite perimeter:
$$\int_E \operatorname{div} X \, \mathrm{d}x = \int_{\partial^* E} X \cdot \nu_E \, \mathrm{d}\mathcal{H}^{n-1},$$
where $\partial^* E$ is the reduced boundary and ν_E the measure-theoretic outer normal; no topological regularity of ∂E is required [6].

3. **Geometric measure theory: Stokes as a definition.** In the theory of currents (de Rham, Federer–Fleming) a k-current is a continuous linear functional T on compactly supported k-forms, and the boundary is defined by $(\partial T)(\alpha) = T(\mathrm{d}\alpha)$, *i.e.* Stokes' formula (M1:15.2) becomes an axiom [7]. Whitney's flat chains and Harrison's chainlets [8] give Stokes' theorem for very irregular (even fractal) domains.

4. **Discrete and numerical versions.** On a simplicial or cell complex, Stokes' theorem holds exactly: the coboundary operator is the transpose of the boundary matrix. This is the foundation of discrete exterior calculus and finite element exterior calculus (Arnold–Falk–Winther [9]), and of structure-preserving discretisations of Maxwell's equations and fluid dynamics that conserve fluxes to machine precision.

5. **Cohomological and topological consequences.** Stokes' theorem underlies de Rham's theorem, Poincaré duality ($\int_M \alpha \wedge \beta$ pairs H^k with H^{n-k} on compact oriented M), degree theory and characteristic classes (via Chern–Weil theory), and the integral laws of electromagnetism and conservation laws in continuum mechanics and general relativity.

6. **Geometric algebra and Clifford analysis.** The vector derivative ∇ and its monogenic functions (Section 17.8) generalise holomorphic function theory to every dimension; the Clifford algebras of Section 1 underlie spin geometry and the Dirac operator, and the geometric-calculus form of Stokes' theorem is the subject of Hestenes–Sobczyk [22]; for spin geometry see [26].

21 Master summary

Prerequisites

Part M1:I; Part M1:II; M1:11 (Tangent and cotangent spaces); 5 (Synthesis I: contraction, duality and the role of the metric); 8 (The exterior derivative in geometric algebra); 9 (Synthesis II: one calculus, three languages); Part III

21.1 One picture

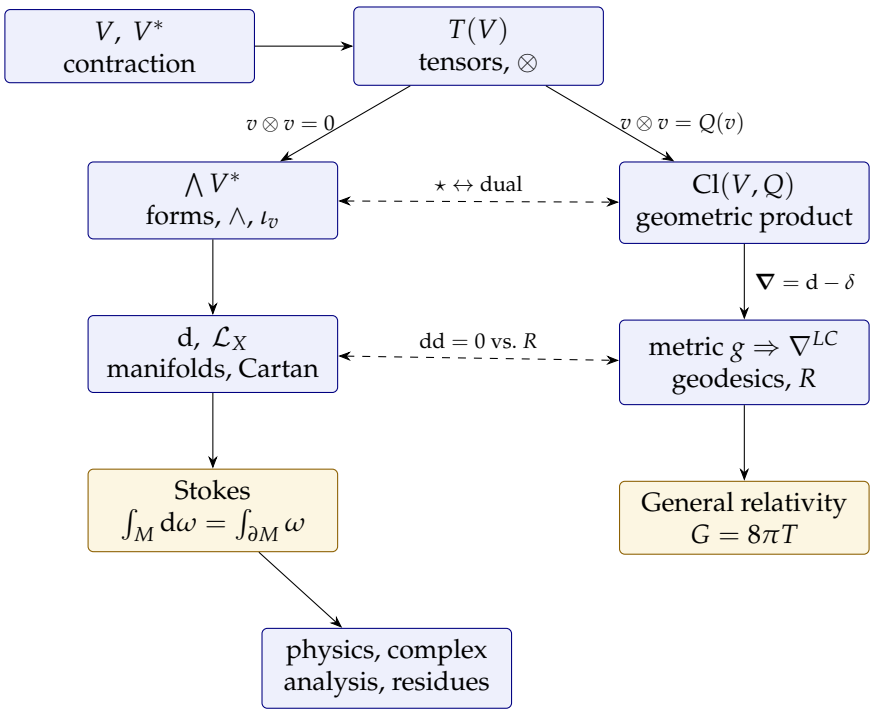


Figure 3: The logical structure of the document. Solid arrows: "is used to build"; dashed: the comparisons made in the synthesis sections.

21.2 The essence of each concept

Concept	Essence
Contraction $\omega(v) = \omega_i v^i$	canonical pairing $V^* \times V \to \mathbb{R}$; needs no metric. First case $\iota_v \omega$ of the interior product.
Exterior product $\alpha \wedge \beta$	antisymmetrised tensor product, a determinant on 1-forms; graded commutative; $\mathrm{d}x$ is the covector dual to $\partial/\partial x$, with $\mathrm{d}x(\partial_x) = 1$.
$\int_\gamma \omega$	contract ω with the velocity, integrate in t: $\int \omega(\dot\gamma)\,\mathrm{d}t = \int_\gamma \gamma^* \omega$. Linearity $\Rightarrow$ reparametrisation invariance and sign change with orientation.
$\int_\phi \alpha$	full contraction with the tangent k-frame, then Lebesgue integration.
Flux $\int_\Sigma \iota_X \mu$	partial contraction of a volume form with a vector field; volume swept per unit time.
Stokes	$\int_M \mathrm{d}\omega = \int_{\partial M} \omega \iff$ d and ∂ are adjoint.

Concept	Essence
Gauss	Stokes applied to $\omega = \iota_X \mu$, using $\mathrm{d}\iota_X\mu = \mathcal{L}_X\mu = \mathrm{div}_\mu X\,\mu$.
Physics	work $\int F$ is intrinsic (force is a covector); the metric appears only as inertia (kinetic energy), as the vacuum (constitutive laws) or via an observer; fluxes are 2-forms, charges 3-forms.
Residues	$f\,\mathrm{d}z$ is closed iff f is holomorphic; $\oint f\,\mathrm{d}z = 2\pi i \sum \mathrm{Res}$ is Stokes on Ω minus discs. Real part = circulation, imaginary part = flux; the residue is the period over a small circle (de Rham pairing).
Geometric algebra	$uv = u \cdot v + u \wedge v$: one product containing the metric and the wedge; $\mathrm{d} \leftrightarrow \boldsymbol{\nabla}\wedge$; one operator $\boldsymbol{\nabla}$ for grad, curl, div; $\boldsymbol{\nabla} F = J$.
Curvature	failure of covariant derivatives to commute: $[\nabla_X, \nabla_Y] - \nabla_{[X,Y]} = R(X,Y)$; the nonzero analogue of $\mathrm{dd} = 0$.

21.3 The toolkit of the geometry of a metric

Object	Definition (formal)	Picture
Dual V^*	linear maps $V \to \mathbb{R}$; $e^i(e_j) = \delta^i_j$	rulers / stacks of planes
Tensor $\mathcal{T}^\ell_k$	multilinear on $V^k \times (V^*)^\ell$; components transform by A (lower), A^{-1} (upper)	coordinate-free machine
$\otimes$	$(f \otimes g)(\cdot,\cdot) = f(\cdot)g(\cdot)$	"machines side by side"
Contraction	plug a covector slot into a vector slot	traces
Metric g	symmetric, non-degenerate $(0,2)$ tensor	lengths, angles, light cones
$g^\flat, g^\sharp$	$v \mapsto g(v,\cdot)$ and inverse	raise/lower indices
Manifold	atlas with C^∞ transitions	locally flat, globally curved
T_pM, T^*_pM	derivations; dual	velocities; measuring devices
$\mathcal{L}_X$	$\frac{\mathrm{d}}{\mathrm{d}\varepsilon}{\phi_\varepsilon}^*\tau$	drag along the flow
∇	axioms (i)–(iii); ${\Gamma^k}_{ij}$	parallel transport
Torsion	$\nabla_X Y - \nabla_Y X - [X,Y]$	twisting (set to 0)
∇^{LC}	unique metric, torsion-free connection; $\Gamma = \frac{1}{2} g^{-1}(\partial g + \partial g - \partial g)$	natural connection of g
Geodesic	$\nabla_{\dot\gamma}\dot\gamma = 0$	straightest path; free fall
Riemann R	$\nabla_X\nabla_Y - \nabla_Y\nabla_X - \nabla_{[X,Y]}$	holonomy; tidal forces
Ricci, s	contractions of R	volume focusing
Geodesic deviation	$D^2\xi = R(\dot\gamma,\xi)\dot\gamma$	neighbouring geodesics

21.4 The dictionary of the algebras

Concept	Formal definition	Picture
Tensor product $V \otimes W$	universal bilinear map; basis $e_i \otimes f_j$	table of numbers $v^i w^j$
Tensor algebra $T(V)$	free associative algebra on V	words in the alphabet V
Exterior algebra $\bigwedge V$	$T(V)/\langle v \otimes v \rangle$	oriented lengths, areas, volumes
Geometric algebra $\mathrm{Cl}(V, Q)$	$T(V)/\langle v \otimes v - Q(v) \rangle$	lengths, angles, areas, rotations in one product
Geometric product	$uv = u \cdot v + u \wedge v$	parallel part + perpendicular part
Contraction $A \rfloor B$	$\langle AB \rangle_{s-r}$	part of B perpendicular to A
Rotor $e^{-\varphi B/2}$	even versor, $\mathsf{R}\widetilde{\mathsf{R}} = 1$	two reflections = rotation in plane B
Dual $A^* = AI^{-1}$	$\mathcal{G}^k \to \mathcal{G}^{n-k}$, $(A \wedge B)^* = A \rfloor B^*$	orthogonal complement; wedge $\leftrightarrow$ meet
Hodge star $\star$	$\alpha \wedge \star\beta = \langle \alpha, \beta \rangle I$	dual up to a sign $(\widetilde{A}I)$
Exterior derivative d	$\mathrm{d} \leftrightarrow \boldsymbol{\nabla}\wedge$; $\mathrm{d}^2 = 0$	curl-like part of $\boldsymbol{\nabla}$; Stokes
Vector derivative $\boldsymbol{\nabla}$	$\boldsymbol{\nabla} = \mathrm{d} - \delta$, $\boldsymbol{\nabla}^2 = -\Delta_H$	grad, div, curl in one operator

21.5 Common mistakes to avoid

1. **Calling the array $\Gamma^k{}_{ij}$ a tensor.** It is not; $\nabla\tau$ and R are tensors.
2. **Mixing up covariant and contravariant.** Test the type of an object by its transformation law: if it uses $\partial\tilde{x}/\partial x$ it is upper-index, if $\partial x/\partial\tilde{x}$ it is lower-index.
3. **Treating the gradient as a vector** without a metric ($\mathrm{d}f$ is a covector; $\nabla f = (\mathrm{d}f)^\sharp$ needs g).
4. **Assuming $\Gamma = 0$ means flat.** Flat means $R = 0$.
5. **Forgetting the Lorentzian signs**: $g(v, v)$ can be negative or zero for $v \neq 0$.
6. **Mixing curvature conventions** between books, **or signature conventions**: mostly plus versus mostly minus (Section 5.3).
7. **Confusing $\mathrm{d}x$ with "a small increment" and $\partial/\partial x$ with a ratio**: the first is a covector, the second depends on the whole chart (Section M1:11.3).
8. **Thinking Stokes needs a metric.** It needs only a smooth structure and an orientation; the metric enters when forms are turned into vector fields (Section 9.4).
9. **Forgetting a factor $k!$** when moving between the two realisations of the wedge (Remark 5.4).
10. **Conflating ∇ and $\boldsymbol{\nabla}$**, or R and the rotor R.

21.6 Suggested path through the references

- *eigenchris, "Tensor Calculus" (YouTube playlist)*: visual explanations of forward/backward transformations, covectors as stacks, tensor products, metric tensor, covariant derivative and Christoffel symbols.

- J. M. Lee, *Introduction to Smooth Manifolds* and *Riemannian Manifolds* [2]: formal proofs of the material of Parts M1:II–III.
- S. Carroll, *Spacetime and Geometry* [21]: physics-oriented derivations; ch. 1–3.
- Doran–Lasenby [23] and Hestenes–Sobczyk [22] for the geometric algebra of Part M1:I and Section 8.
- M. Spivak, *Calculus on Manifolds* [1], for the cleanest proof of Stokes' theorem.

22 Topology: homotopy, homology and de Rham cohomology

TODO

TODO(M2/ch22): this chapter has no content yet. Its place in the reading order is listed in `CHAPTER_ORGANIZATION.md`.

Prerequisites: M1:12 (Tensor fields and differential forms); M1:15 (Stokes' theorem as it is understood today); TODO(M2/ch06)

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[3] R. Bott and L. W. Tu, *Differential Forms in Algebraic Topology*, Springer GTM 82, 1982.

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